

CLASSIFICATION OF INTEGRAL MODULAR DATA UP TO RANK 13

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ABSTRACT. This paper classifies the modular data of integral modular fusion categories up to rank 13, and integral half-Frobenius fusion rings up to rank 12. We establish that every perfect case within these bounds is trivial. Furthermore, we refine the non-pointed odd-dimensional modular data at ranks below 25 to exactly three items, all of rank 17, FPdim 225, and type $[[1,3],[3,8],[5,6]]$, filling existing literature gaps. For rank 25, we narrow the perfect case to three open types.

Our core insight is that Egyptian fractions, typically used to list possible types, can be chosen with squared denominators. We develop several type criteria as initial filters. To construct the fusion rings, we solve dimension and associativity equations utilizing custom-built features in Normaliz. S-matrices are generated by self-transposing the character table, and T-matrices are derived by solving the Anderson-Moore-Vafa equations, concluding with the verification of extended modular data axioms.

From rank 13 onward, types are restricted by modular-specific properties involving universal grading, congruence representations of the modular group, and Galois action. This establishes critical arithmetic constraints: up to rank 21, a prime divisor of the global FPdim cannot exceed the rank, and up to rank 15 (non-pointed case), it cannot exceed half the rank. Ultimately, we reduce the rank 14 classification to 35 possible types, 8 of which are non-perfect.

1. INTRODUCTION

In this paper, we assume that all fusion categories are defined over the complex field. The concept of an integral modular fusion category has been extensively studied, as detailed in the references at the beginning of [15]. In [8], they have been classified up to rank 6 (all pointed), with Egyptian fractions playing a crucial role. The approach that enables us to extend this classification up to rank 13 in our work hinges on the observation that it is sufficient to consider Egyptian fractions with squared denominators. This restriction significantly reduces the combinatorial complexity. To illustrate this point, consider that the number of Egyptian fractions (summing to 1) of length $n = 1, 2, \dots, 8$ is 1, 1, 3, 14, 147, 3462, 294314, 159330691, respectively (as per [52]). In contrast, when limited to squared denominators, the counts are 1, 0, 0, 1, 0, 1, 1, 4, respectively (refer to [1]).

We begin by recalling the concept of a fusion ring and its fundamental results in §2.1, with reference to [22, Chapter 3]. As defined in [25], a fusion ring $\mathcal{F}$ is termed *s-Frobenius* if for every basic element b , the ratio $\text{FPdim}(\mathcal{F})^s / \text{FPdim}(b)$ is an algebraic integer. According to [22, Proposition 8.14.6], the Grothendieck ring of a modular fusion category is 1/2-Frobenius (denoted *half-Frobenius* in the rest of the paper). Consider $\mathcal{F}$ to be an integral half-Frobenius fusion ring with a basis $\{b_1, \dots, b_r\}$, FPdim D , and type $[d_1, \dots, d_r]$, where $1 = d_1 \leq d_2 \leq \dots \leq d_r$ and $d_i = \text{FPdim}(b_i)$. Thus d_i^2 is a divisor of D , for all i . There exists a unique square-free integer q such that $D = qs^2$, implying that each d_i is a divisor of s . Let s_i denote the positive integer s/d_i . Given that $D = \sum_{i=1}^r d_i^2$, we arrive at the following representation of q as an Egyptian fraction with squared denominators $q = \sum_{i=1}^r 1/s_i^2$. We have classified all such Egyptian fractions up to $r = 13$ using SageMath, as will be discussed in §4, where a method to constrain to $q \leq r/4$ is also described (Proposition 4.1). Since $s_1 = s$, we have $d_i = s_1/s_i$, and we may assume that s_i is a divisor of s_1 , for all i . As detailed in §7.1, this leads us to consider only 9025 types up to rank 13.

The subsequent phase entails implementing new criteria for identifying a type that emerges from a fusion ring, as delineated in §5. The proof of these criteria predominantly relies on modular arithmetic and serves to rule out approximately 62% of the types up to rank 13.

To address the remaining types, we classify all possible fusion data $(N_{i,j}^k)$, as defined in §2.1, for each type (d_i) , utilizing our fusion ring solver described in §6. We begin by reducing the number of variables, leveraging the Unit axiom of fusion data and the Frobenius reciprocity. The main challenge, denoted as *patching*, involves integrating the associativity equations $\sum_s N_{i,j}^s N_{s,k}^t = \sum_s N_{j,k}^s N_{i,s}^t$ (which are non-linear) as efficiently as possible into the (linear) solving process of the dimension equations $d_i d_j = \sum_k N_{i,j}^k d_k$, which are positive linear Diophantine equations. This approach was implemented using Normaliz [11], on which we developed new features dedicated to the classification of fusion rings, as explained in §6.1, and for more details, see Appendix H of the manual [12].

This step culminates in a classification of all the half-Frobenius integral fusion rings up to rank 12, tallying exactly 10628 instances derived from 71 types, and proves the absence of any non-trivial perfect¹ integral half-Frobenius fusion rings up to rank 12 (see §7). We can limit our attention to commutative fusion rings since a modular fusion category, being braided, possesses a commutative Grothendieck ring (although our classification encompasses 213 noncommutative fusion rings as well; see §7).

From rank 13 onward, the types were further restricted by additional properties coming from more advanced results on modular fusion categories, see §4 and §8. This adjustment was necessary because we encountered computational limits for classifying all half-Frobenius fusion rings. Consequently, the result became less general at the fusion ring level compared to what we get up to rank 12. In the non-perfect case, we applied specific *universal grading* techniques, as discussed in §8.1 and based on [38, Proposition VI.2]. Additionally, we explored congruence representations of the modular group (see §8.2) and Galois actions (see §8.3). In particular, we provide two proofs of Theorem 8.4, a folklore² result: For any prime factor p of the dimension norm of a modular fusion category with rank r , it holds that $p \leq 2r + 1$. The shorter proof applies [21, Theorem II (iii)]. This inequality is optimal, and the examples for which the equality holds are classified in [41]. Regarding the integral case, discussions with Eric Rowell and Andrew Schopieray indicated that the rank r can be substituted with the multiplicity m of a certain basic FPdim, leading to the inequality $p \leq 2m + 1$, see Theorem 8.5, required to exclude some hard types at rank 13. We have a stronger version:

Theorem 8.7. *For an integral modular fusion category, let S be the set of odd prime factors of the global FPdim. There is a partition (S_i) of S , and multiplicities (m_i) of some distinct basic FPdims such that*

$$m_i \geq \frac{1}{2} \operatorname{lcm}_{p \in S_i} (p - 1).$$

It was crucial for proving Proposition 12.8 for the odd-dimensional case, and also the following theorem from §9.2:

Theorem 1.1. *Let $\mathcal{C}$ be an integral modular fusion category of rank r . For any prime factor p of $\operatorname{FPdim}(\mathcal{C})$, the following bounds hold:*

- (1) *If $r \leq 21$, then $p \leq r$.*
- (2) *If $r \leq 15$ and $\mathcal{C}$ is non-pointed, then $p \leq r/2$.*

In the non-perfect case, the above upper bounds can be improved to $r \leq 24$ and $r \leq 16$, respectively.

These inequalities are conjectured to hold without rank restriction in §9.2. These conjectures are proved for $\mathcal{Z}(\operatorname{Rep}(G))$, across all finite groups G , in §9.1, among other results.

At rank 13, we obtained 28998 fusion rings from 10 types. The next step is to classify all possible modular data related to the fusion rings found up to rank 13. The definition of modular data we employ (refer to §2.2) is informed by the key attributes of a modular fusion category, specifically a pseudo-unitary one, as our research is centered on the integral case (see [22, Proposition 9.6.5]).

First, we examine the S -matrices: for a given commutative fusion ring, we take its eigentable (as defined in Definition 2.7) and consider it as a matrix, retaining only those with cyclotomic elements—such fusion rings are termed *cyclotomic*. If suitable renormalization and permutation yield a self-transpose matrix (detailed in §3.1), we call the fusion ring as *self-transposable*; if not, it is dismissed. From this, we infer that there are precisely 69 self-transposable, cyclotomic, half-Frobenius, integral fusion rings up to rank 12, originating from 27 types, which is fewer than 0.7% of the 10628 identified in the initial stage. At rank 13 (more restricted), we reduced to 10 non-pointed fusion rings from 2 types.

Moving on to the T -matrices: for the fusion rings that remain, we solve the Anderson-Moore-Vafa equations (see §2.2) in the $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$. We preserve only those S - and T -matrices that satisfy all the conditions of Definition 2.11. The use of our so-called *magic criterion* was pivotal for several big cases and could lead to an interesting theoretical reformulation, see Question 3.2. Ultimately, we arrive at 19 + 64 modular data, derived from 5 + 18 fusion rings and 3 + 13 types (non-pointed + pointed).

Remark 1.2. Every pointed modular fusion category corresponds to a metric group (G, q) —a finite Abelian group G equipped with a non-degenerate quadratic form $q : G \rightarrow \mathbb{C}^*$, represented by the T -matrix, as described in [22, §8.4].

The modular data (MD) mentioned in [3, §I] encompass S - and T -matrices, central charge, fusion data, and second Frobenius-Schur indicators for the non-pointed case. For the pointed case, however, it includes only the T -matrices. The following theorem, proved in §10, provides a concise overview:

Theorem 1.3. *There are 19 MD of non-pointed integral modular fusion categories up to rank 13, given by:*

- Rank 8, FPdim 36, type [1, 1, 2, 2, 2, 2, 3, 3]:

¹Recall that *perfect* means exactly one basic element with $\operatorname{FPdim} = 1$. A non-pointed simple fusion ring is perfect; *pointed* means all basic elements have $\operatorname{FPdim} = 1$, and *simple* means no non-trivial proper fusion subrings.

²Throughout the paper, the term *folklore* is used solely to indicate that the result is well-known to experts but not explicitly stated in the literature. It does not imply anything beyond that.

- 6 MD with central charge $c = 0$ from $\mathcal{Z}(\text{Vec}_{S_3}^\omega)$, see [29],
- 2 MD with $c = 4$ from $(C_3^2 + 0)^{C_2}$, see [28, point (b) on page 983].
- Rank 10, FPdim 36, type $[1, 1, 1, 2, 2, 2, 2, 2, 3]$:
 - 3 MD with $c = 4$, from $SU(3)_3$, its complex conjugate and a zesting, see [19, §6.3.1].
- Rank 11, FPdim 32, type $[1, 1, 1, 1, 2, 2, 2, 2, 2, 2]$:
 - 8 MD with $c = \pm 1$, from $SO(8)_2$, a zesting and their Galois conjugates, see §3.3.

There are 64 modular data of pointed modular fusion categories up to rank 13. Their number per group G is in Table 1.

G	C_1	C_2	C_3	C_2^2	C_4	C_5	C_6	C_7	C_2^3	$C_2 \times C_4$	C_8	C_3^2	C_9	C_{10}	C_{11}	$C_2 \times C_6$	C_{12}	C_{13}
#MD	1	2	2	5	4	2	4	2	4	4	4	2	2	4	2	10	8	2

TABLE 1. Number of modular data for pointed modular fusion categories of rank at most 13

We found no other integral modular data up to rank 13 apart from the categorifiable ones listed above.

Question 1.4. *Is there a modular data without categorification?*

Note that [38] presents five intriguing non-integral candidates of ranks 11 and 12 (see [38, I.C]), which are relevant to Question 1.4; and it also agrees with³ Theorem 1.3 up to rank 12. This theorem yields the following consequence:

Theorem 1.5. *Every perfect integral modular fusion category up to rank 13 is trivial.*

In fact, we obtained the following more general result within the context of fusion rings up to rank 12:

Theorem 1.6. *Every perfect integral half-Frobenius fusion ring up to rank 12 is trivial.*

The proof of Theorem 1.6 for ranks up to 9 is straightforward, following the list mentioned in §7.1 combined with an extended version of the Nichols-Richmond theorem applied to fusion rings, as detailed in the proof of [42, Theorem 11]. This is due to the consistent presence of a non-trivial basic element with $\text{FPdim} \leq 2$. However, proving the theorem for ranks up to 12 necessitates the employment of type criteria, as discussed in §5, and the use of a fusion ring solver, elaborated in §6.

It should be noted that the Drinfeld center of the representation category of any non-Abelian finite simple group G —and, more broadly, any centerless perfect group—is a perfect (though not simple) integral modular fusion category denoted as $\mathcal{Z}(\text{Rep}(G))$ with $\text{FPdim} = |G|^2$. For further information, see [13, §11.1]. Thus, the Grothendieck ring of $\mathcal{Z}(\text{Rep}(A_5))$, of rank 22 and type $[[1, 1], [3, 2], [4, 1], [5, 1], [12, 10], [15, 4], [20, 3]]$, constitutes a perfect integral half-Frobenius fusion ring. Consequently, Theorem 1.6 cannot be extended to all ranks; however, it remains an open question whether its simple version can be:

Question 1.7. *Is there a non-pointed simple integral half-Frobenius fusion ring?*

A negative response to Question 1.7 would imply a negative answer to the renowned [24, Question 2] in the simple case, due to a result in [33], which states that every simple integral fusion category is weakly group-theoretical if and only if every simple integral modular fusion category is pointed. With this in mind, we propose the following question:

Question 1.8. *Is there a non-pointed simple integral modular fusion category?*

For further insights into Question 1.8 at the fusion ring level, [43, Corollary 6.16] adds a constraint: the absence of any prime-power basic FPdim. It is worth noting that Theorem 1.6 cannot be generalized to all ranks, even with this added constraint. This is because the Grothendieck ring of $\mathcal{Z}(\text{Rep}(A_7))$, which is a perfect integral half-Frobenius fusion ring of rank 74 and type

$$[[1, 1], [6, 1], [10, 2], [14, 2], [15, 1], [21, 1], [35, 1], [70, 9], [105, 4], [210, 20], [280, 9], [360, 14], [504, 5], [630, 4]],$$

satisfies this constraint (but consider Question 7.2). If necessary, Question 1.7 could be refined to include this constraint and the property of commutativity, and even more advanced constraints from §8.

Using similar methods, in §11 we reduce the rank-14 classification to just 35 possible types, 27 of which are perfect—each having an FPdim whose set of prime divisors is exactly $\{2, 3, 5, 7\}$; see also Remark 11.4. Additionally, in conjunction with [15, Remark 4.3] and [43, Corollary 6.16], we are able to demonstrate the following (refer to §12):

Theorem 1.9. *There are three MD of non-pointed odd-dimensional modular fusion categories at rank below 25:*

- Rank 17, FPdim 225, type $[1, 1, 1, 3, 3, 3, 3, 3, 3, 3, 5, 5, 5, 5, 5, 5]$:

³Our classification up to rank 12 appeared in an earlier version as arXiv:2302.01613v3 in May 2023, three months before the release of [38] as arXiv:2308.09670 in August 2023. Our collaboration began in September 2021 at <https://mathoverflow.net/q/403397/34538>.

– 3 MD with central charge $c = 4$ from 2 fusion rings (see [3, §II] for the details).

Remark 1.10. Theorem 1.9 does not align with [6, Theorem 4.2, proof of Case (viii) $\text{FPdim}(\mathcal{C}_{pt}) = p$] as well as [15, Theorem 6.3 (b), proof of Case $|\mathcal{G}(\mathcal{C})| = 3$]. But Remark 12.6 points out gaps in these proofs. Following our paper, [15] was corrected on arXiv, and [26] introduces modular category models for these new MD.

Finally, this paper narrows down the possible rank 25 odd-dimensional perfect types to 3 ones, see Proposition 12.8.

CONTENTS

1. Introduction	1
2. Fusion Data and Modular Data	4
2.1. Fusion Data	4
2.2. Modular Data	6
3. From Fusion Data to Modular Data	7
3.1. S-matrix	7
3.2. T-matrix	7
3.3. Models for Rank 11	7
4. Egyptian Fractions with Squared Denominators	8
5. Type Criteria	9
5.1. Small Perfect Type	9
5.2. Gcd Criterion	10
5.3. Type Test	10
5.4. Local Criterion	11
6. Enhanced Fusion Ring Solver Using Normaliz	12
6.1. Normaliz User Interface for Fusion Rings	12
6.2. Normaliz and its Approach to Fusion Rings	13
6.3. Full Version	14
6.4. Dimension Partition Version	15
7. Half-Frobenius Integral Fusion Rings up to Rank 12	16
7.1. List of Possible Types	16
7.2. List of Fusion Rings	16
7.3. Commutative, Cyclotomic and Self-Transposable	17
8. Advanced Results on Modular Fusion Categories	17
8.1. Universal Grading	17
8.2. Congruence Representation	18
8.3. Galois Action	19
9. Stronger Arithmetic Constraints	20
9.1. Rank of $\mathcal{Z}(\text{Rep}(G))$	20
9.2. Integral Modular Fusion Categories	22
10. Proof Up To Rank 13	23
11. About Ranks 14 and 15	24
12. The Odd-Dimensional Case	24
A. Proof by Geoff Robinson	26
About the Supplementary Material	27
Acknowledgments	28
References	28

2. FUSION DATA AND MODULAR DATA

In this section, we review the concepts of fusion data and modular data, along with the essential results. For further details, we refer the reader to [22].

2.1. Fusion Data. The concept of fusion data expands upon the idea of a finite group.

Definition 2.1. *Fusion data* consist of a finite set $\{1, 2, \dots, r\}$ with an involution $i \mapsto i^*$, and nonnegative integers $N_{i,j}^k$ satisfying the following conditions for all i, j, k, t :

- (Associativity) $\sum_s N_{i,j}^s N_{s,k}^t = \sum_s N_{j,k}^s N_{i,s}^t$,
- (Unit) $N_{1,i}^j = N_{i,1}^j = \delta_{i,j}$,

- (Dual) $N_{i^*,j}^1 = N_{j,i^*}^1 = \delta_{i,j}$,
- (Anti-involution) $N_{i,j}^k = N_{j^*,i^*}^{k^*}$.

Note that $1^* = 1$. We may represent the fusion data simply as $(N_{i,j}^k)$.

Proposition 2.2 (Frobenius Reciprocity). *For all i, j, k , $N_{i,j}^k = N_{k,j^*}^i = N_{k^*,i}^{j^*} = N_{j^*,i^*}^{k^*} = N_{j,k^*}^i = N_{i^*,k}^j$.*

Proof. Starting with (Associativity) and setting $t = 1$, we have $\sum_s N_{i,j}^s N_{s,k}^1 = \sum_s N_{j,k}^s N_{i,s}^1$. Applying (Dual), we get $\sum_s N_{i,j}^s \delta_{s,k^*} = \sum_s N_{j,k}^s \delta_{s,i^*}$. Consequently, $N_{i,j}^{k^*} = N_{j,k}^i$. Substituting k^* with k , we obtain $N_{i,j}^k = N_{j,k^*}^i$, which equals N_{k,j^*}^i by (Anti-involution). The proposition follows by iterating the equality $N_{i,j}^k = N_{k,j^*}^i$ and (Anti-involution). $\square$

Remark 2.3. We can construct data that satisfy the first three axioms of Definition 2.1 but not the fourth, proving it is not superfluous. However, (Unit) is redundant when combined with the other axioms, as it is not utilized in the proof of Proposition 2.2. Taken together, (Dual) and (Frobenius Reciprocity) trivially imply (Unit).

A *fusion ring* $\mathcal{R}$ is a free $\mathbb{Z}$ -module equipped with a finite basis $\mathcal{B} = \{b_1, \dots, b_r\}$ and a fusion product defined by

$$b_i b_j = \sum_k N_{i,j}^k b_k,$$

where $(N_{i,j}^k)$ constitutes fusion data, and a $*$ -structure given by $b_i^* := b_{i^*}$. The four axioms for fusion data translate to the following for all i, j, k :

- $(b_i b_j) b_k = b_i (b_j b_k)$,
- $b_1 b_i = b_i b_1 = b_i$,
- $\tau(b_i b_j^*) = \delta_{i,j}$,
- $(b_i b_j)^* = b_j^* b_i^*$,

where $\tau(x)$ is the coefficient of b_1 in the decomposition of $x \in \mathcal{R}$. Consequently, $\mathcal{R}_{\mathbb{C}} := \mathcal{R} \otimes_{\mathbb{Z}} \mathbb{C}$ becomes a finite-dimensional unital $*$ -algebra, with τ extending linearly to a trace (i.e., $\tau(xy) = \tau(yx)$) and an inner product defined by $\langle x, y \rangle := \tau(xy^*)$. Here, $\langle x, b_i \rangle$ is the coefficient of b_i in the decomposition of x .

Theorem 2.4 (Frobenius-Perron Dimension Theorem [23, §8]). *Given a fusion ring $\mathcal{R}$ with basis $\mathcal{B}$ and the corresponding finite-dimensional unital $*$ -algebra $\mathcal{R}_{\mathbb{C}}$, there exists a unique $*$ -homomorphism $d : \mathcal{R}_{\mathbb{C}} \rightarrow \mathbb{C}$ such that $d(\mathcal{B}) \subset \mathbb{R}_{>0}$.*

The value $d(b_i)$, known as the *Frobenius-Perron dimension* of b_i , is denoted as $\text{FPdim}(b_i)$ or simply d_i . This is referred to as a *basic* FPdim. The sum $\sum_i d_i^2$ is called the Frobenius-Perron dimension of $\mathcal{R}$, or *global* FPdim, and is denoted $\text{FPdim}(\mathcal{R})$. The sequence $[d_1, d_2, \dots, d_r]$ is called the *type* of $\mathcal{R}$. A fusion ring $\mathcal{R}$ is described as:

- *Frobenius* (or 1-Frobenius, or of Frobenius type) if $\frac{\text{FPdim}(\mathcal{R})}{\text{FPdim}(b_i)}$ is an algebraic integer for all i ,
- *integral* if $\text{FPdim}(b_i)$ is an integer for all i ,
- *pointed* if $\text{FPdim}(b_i) = 1$ for all i ,
- *commutative* if $b_i b_j = b_j b_i$ for all i, j , meaning $N_{i,j}^k = N_{j,i}^k$.

The *multiplicity* of $\mathcal{R}$ is the maximum value among $N_{i,j}^k$, and its *rank* is r , the size of the basis.

Remark 2.5. Fusion data enable a representation of its corresponding fusion ring. Consider the matrices $M_i = (N_{i,j}^k)_{k,j}$. By (Associativity) in Definition 2.1, we verify that $M_i M_j = \sum_k N_{i,j}^k M_k$. Additionally, M_1 is the identity matrix, and Frobenius Reciprocity ensures that the adjoint matrix M_i^* is M_{i^*} . According to Frobenius-Perron Theorem, the operator norm $\|M_i\|$ equals $\text{FPdim}(b_i)$.

Remark 2.6. The concept of fusion data is a combinatorial reformulation of the fusion ring notion, so any property applicable to a fusion ring is also applicable to its fusion data.

Definition 2.7 (Eigentable). Given commutative fusion data $(N_{i,j}^k)$, consider the corresponding fusion matrices $M_i = (N_{i,j}^k)_{k,j}$. The commutativity and the property that $M_i^* = M_{i^*}$ render these matrices normal and thus simultaneously diagonalizable. Let (D_i) denote their simultaneous diagonalization, where $D_i = \text{diag}(\lambda_{i,j})$. We can select $\lambda_{i,1} = \|M_i\| = d_i$. The matrix $(\lambda_{i,j})$ is termed the *eigentable* (or character table) of the fusion data, and the values $c_j := \sum_i |\lambda_{i,j}|^2$ are called the *formal codegrees*.

Lemma 2.8. *Let $M \in M_n(\mathbb{Z}_{\geq 0})$. The matrix M is a permutation matrix if and only if $\|M\| = 1$.*

Proof. Consider an orthonormal basis $\{e_1, \dots, e_n\}$ for which the entries of M are non-negative integers. If M is not a permutation matrix, then one of the following cases must occur:

- (0) there exists i for which $M e_i = 0$,
- (1) there exist i, j such that $\langle M e_i, e_j \rangle > 1$,
- (2) there exist i, j, k with $j \neq k$, such that $\langle M e_i, e_j \rangle = \langle M e_i, e_k \rangle = 1$,

(3) there exist i, j, k with $i \neq j$, such that $Me_i = Me_j = e_k$.

However, case (0) implies $\|Me_i\|/\|e_i\| = 0$, while case (1) leads to $\|Me_i\|/\|e_i\| > 1$. In case (2), it follows that $\|Me_i\|/\|e_i\| \geq \sqrt{2}$. Likewise, case (3) implies $\|M(e_i + e_j)\|/\|e_i + e_j\| = \sqrt{2}$. Each of these cases indicates that $\|M\| > 1$. Conversely, if M is a permutation matrix, it trivially follows that $\|M\| = 1$. $\square$

Corollary 2.9. *For two basic elements x, y of a fusion ring with $\text{FPdim}(x) = 1$, both xy and yx are basic elements, and $\text{FPdim}(xy) = \text{FPdim}(yx) = \text{FPdim}(y)$.*

Proof. This follows directly from Remark 2.5, Lemma 2.8, and the fact that FPdim is a ring homomorphism. $\square$

Corollary 2.10. *A fusion ring is pointed if and only if its basis forms a finite group under the fusion product.*

2.2. Modular Data. Broadly speaking, modular data refers to a fusion data together with two matrices, S and $T = (t_{i,j})$, that generate a projective representation of the modular group $\text{SL}(2, \mathbb{Z})$. To provide a more detailed description, we draw upon [37, Theorem 2.1] and [22, §8.13, §8.18]. Let $\mathbf{i}$ be the imaginary unit.

Definition 2.11. Given a fusion ring $\mathcal{R}$ of rank r , type $[d_1, \dots, d_r]$, and fusion data $(N_{i,j}^k)$, let $\mathbf{d} := \text{FPdim}(\mathcal{R})$ and $\zeta_n := \exp(2\pi\mathbf{i}/n)$. A (pseudounitary) *modular data* for $\mathcal{R}$ consists of two matrices $S, T \in M_r(\mathbb{C})$ satisfying:

- S and T are symmetric, T is unitary and diagonal with $T_{1,1} = 1$, $S_{1,i} = d_i$ for all i , and $SS^* = \mathbf{d}\mathbf{1}$.
- Verlinde formula: $N_{i,j}^k = \frac{1}{\mathbf{d}} \sum_l \frac{S_{li} S_{lj} \overline{S_{lk}}}{d_l}$.
- Twist: let θ_i be $T_{i,i}$, then $\sum_k N_{i,j}^k \theta_k d_k = \theta_i \theta_j S_{i,j}$.
- Ribbon structure: $\theta_i = \theta_{i^*}$ (see Remark 2.13).
- Central charge: $p_{\pm} := \sum_{i=1}^r d_i^2 (\theta_i)^{\pm 1}$. The ratio p_+/p_- is a root of unity, and $p_+ = \sqrt{\mathbf{d}} \zeta_8^c$ for some rational number c , referred to as the **central charge**, determined modulo 8.
- The matrices S and T afford a projective representation of $\text{SL}(2, \mathbb{Z})$: we have $(ST)^3 = p_+ S^2$, $\frac{S^2}{\mathbf{d}} = C$, $C^2 = \mathbf{1}$, where C is the permutation matrix associated with the involution $i \rightarrow i^*$ and satisfies $\text{Tr}(C) > 0$.
- Cauchy theorem: the set of distinct prime factors of $\text{ord}(T)$ is identical to the distinct prime factors of $\text{norm}(\mathbf{d})$, where $\text{norm}(x)$ denotes the product of the distinct Galois conjugates of the algebraic number x .
- Cyclotomic integers: for all i, j , the elements $S_{i,j}$, $S_{i,j}/d_j$ and $T_{i,i}$ are cyclotomic integers. The conductor of $S_{i,j}$ divides $\text{ord}(T)$, which in turn divides $\mathbf{d}^{5/2}$, and there exists j such that $S_{i,j}/d_j \in \mathbb{R}_{\geq 1}$, for all i .
- Frobenius-Schur indicators: for every i and for all $n \geq 1$, the sum $\nu_n(i) := \frac{1}{\mathbf{d}} \sum_{j,k} N_{j,k}^i (d_j \theta_j^n) (\overline{d_k \theta_k^n})$ is a cyclotomic integer with a conductor that divides both n and $\text{ord}(T)$. Additionally, $\nu_1(i) = \delta_{i,1}$ and $\nu_2(i) = \pm \delta_{i,i^*}$.
- Anderson-Moore-Vafa equations: $T_{i,i} = e^{2\pi i t_i}$, and $\forall i, j, k, l$, the following equation holds in the $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$:

$$\left(\sum_{p=1}^r N_{i,j}^p N_{p,k}^l \right) (t_i + t_j + t_k + t_l) = \sum_{p=1}^r \left(N_{i,j}^p N_{p,k}^l + N_{i,k}^p N_{j,p}^l + N_{j,k}^p N_{i,p}^l \right) t_p.$$

The **topological spin** of the i -th basic element is the representative $s_i \in (-1/2, 1/2]$ of $t_i \in \mathbb{Q}/\mathbb{Z}$.

We could question the necessity of each component in Definition 2.11, particularly whether the Anderson-Moore-Vafa equations can be inferred from the other assumptions.

Remark 2.12. The Verlinde formula, in conjunction with results from [33, §2], implies that the fusion ring $\mathcal{R}$ is commutative. Together with S symmetric and the identity $SS^* = \text{FPdim}(\mathcal{R})\mathbf{1}$, it can be deduced that $\mathcal{R}$ is self-transposable (as discussed in §3.1). Moreover, according to the proof presented in [22, Proposition 8.14.6], $\mathcal{R}$ is also half-Frobenius.

Remark 2.13. A modular tensor category $\mathcal{C}$ possesses a ribbon structure, which means that the twist $\theta \in \text{Aut}(\text{id}_{\mathcal{C}})$ satisfies the condition $(\theta_X)^* = \theta_{X^*}$ for every object X within $\mathcal{C}$. Let (X_i) represent the set of simple objects (up to isomorphism) within $\mathcal{C}$. Schur's lemma guarantees that $\theta_{X_i} = \theta_i \text{id}_{X_i}$, where the scalar θ_i is consistent with the one described in Definition 2.11. Owing to the ribbon structure, we deduce the following:

$$\theta_{i^*} \text{id}_{X_{i^*}} = \theta_{X_{i^*}} = (\theta_{X_i})^* = (\theta_i \text{id}_{X_i})^* = \theta_i (\text{id}_{X_i})^* = \theta_i \text{id}_{X_{i^*}}.$$

From this, it follows that $\theta_{i^*} = \theta_i$ for all simple objects X_i .

This paper primarily concerns integral fusion categories. In this setting, $\mathbf{d}$ is an integer, so $\text{norm}(\mathbf{d}) = \mathbf{d}$. These categories are pseudounitary and therefore spherical (see [22, Proposition 9.5.1]). In contexts that are not pseudounitary, Definition 2.11 would require modifications (as suggested in [37, Theorem 2.1]) because the equalities $S_{1,i} = \text{FPdim}(b_i)$, for all i , may not be valid. Strictly speaking, the assumption arising from these last equalities should be referred to as *positive*, which lies between *pseudounitary* and *unitary*. Recall that [22, Proposition 9.5.1] states that a pseudounitary fusion category admits a positive structure, but whether it always admits a unitary structure remains an open problem (see [22, page 284]).

It should be noted that the definition of modular data provided here is so stringent that, as of now, no instances exist that lack a categorification, leading to Question 1.4.

3. FROM FUSION DATA TO MODULAR DATA

This section elucidates the classification of all potential modular data associated with a given set of fusion data. Initially, we may consider the fusion data to be commutative and half-Frobenius (refer to Remark 2.12).

3.1. S-matrix. Consider a commutative fusion data $(N_{i,j}^k)$ of rank r , eigentable $(\lambda_{i,j})$, and formal codegrees (c_j) as defined in Definition 2.7. The objective here is to identify all permutations q of the set $\{1, \dots, r\}$ such that:

- $q(1) = 1$,
- $d_{q(i)} = d_i$ for all i ,
- The matrix $S = (\sqrt{c_1/c_j} \lambda_{i,q(j)})$ is symmetric (i.e. self-transpose).

Remark 3.1. The symmetric requirement implies that

$$\sqrt{c_1/c_j} = \sqrt{c_1/c_j} \lambda_{1,q(j)} = \sqrt{c_1/c_1} \lambda_{j,q(1)} = d_j,$$

hence we can infer that $c_1/c_j = d_j^2$ for all j , as shown in [45, Example 2.9].

If such a permutation q exists (Remark 3.1 can serve as an effective necessary condition), the fusion data are referred to as *self-transposable*. This property is exceedingly rare, rendering this step a potent sieve. Using the Verlinde formula, one can reconstruct the fusion data from S . It is important to note that we need only consider *cyclotomic* fusion data, i.e. whose eigentable entries are all cyclotomic.

3.2. T-matrix. For the remaining fusion rings $\mathcal{R}$ with fusion data $(N_{i,j}^k)$, we address the Anderson-Moore-Vafa equations:

$$\left(\sum_{p=1}^r N_{i,j}^p N_{p,k}^l \right) (t_i + t_j + t_k + t_l) = \sum_{p=1}^r \left(N_{i,j}^p N_{p,k}^l + N_{i,k}^p N_{j,p}^l + N_{j,k}^p N_{i,p}^l \right) t_p$$

within the $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$. For each valid solution $t = (t_i) \in (\mathbb{Q}/\mathbb{Z})^r$, if any, the corresponding T -matrix is $\text{diag}(e^{2\pi i t_i})$.

The solutions to the aforementioned equations are determined using the following method: Initially, the matrix reformulation is represented as $At = 0$, where A is an $m \times n$ matrix over $\mathbb{Z}$ with $m = r^4$ and $n = r$. Subsequently, the Smith normal form is employed, denoted as $D = UAV$, in which U and V are invertible matrices over $\mathbb{Z}$ of sizes $m \times m$ and $n \times n$, respectively, and D is a diagonal $m \times n$ matrix $(\alpha_i \delta_{i,j})$, where the integer α_i is divisible by α_{i+1} for all $i < r$, and $\delta_{i,j}$ is the Kronecker delta. The solutions to $Dx = 0$ are precisely represented by the vectors (k_i/α_i) , where $0 \leq k_i < \alpha_i$. Consequently, we have $U^{-1}DV^{-1}t = 0$, which simplifies to $DV^{-1}t = 0$. Therefore, the solutions can be expressed as $t = Vx$.

A complete listing of potential T -matrices requires considering all vectors (k_i/α_i) , where $0 \leq k_i < \alpha_i$. As a result, there are $p = \prod_i \alpha_i$ possible combinations. This task remains manageable up to rank 11. However, in certain case of rank above, the value of p becomes too large. But a miraculous circumstance arises (referred to as the **magic criterion**): for all such cases, if one abstractly considers the T -matrix with variables (k_i) , then for every determined S -matrix S in §3.1, the abstract product $(ST)^3$ consistently exhibits a zero where it should not, specifically at an entry (i, i^*) for some i . This is because $(ST)^3/p_+ = S^2 = \mathbf{d}C$, where C is the duality matrix (realizing the involution $i \rightarrow i^*$, and thus $C_{i,i^*} = 1$, non-zero), as defined in Definition 2.11. The function **MagicCriterion** (also covered by the function **STmatrix**) can verify this.

Question 3.2. *Can the aforementioned magic criterion be reformulated at the level of fusion data?*

3.3. Models for Rank 11. This subsection benefited from helpful discussions with César Galindo and Eric Rowell. Among the eight modular data described in [3, §I.1.5], the last one corresponds exactly to the modular fusion category of finite-dimensional representations of the affine Kac-Moody algebra of type D_4 at level 2 (see [32]), also known as $SO(8)_2$. This can be checked using [27], or a recent version of SageMath [51]:

```
sage: D42 = FusionRing("D4", 2) # SO(8)_2
sage: len(D42.basis())
11 # Rank
sage: [(b.q_dimension()) for b in D42.basis()]
[1, 2, 2, 2, 2, 1, 2, 2, 1, 2, 1] # Type
sage: [x.twist()/2 for x in D42.basis()]
[0, 7/16, 3/4, 7/16, 7/16, 0, 15/16, 15/16, 0, 15/16, 0] # Topological spins
```

We need the factor `x.twist()/2` because SageMath adopts the convention $e^{\pi i t_i}$, whereas we use $e^{2\pi i t_i}$. Up to permutation and modulo 1, we recover the expected topological spins:

$$[0, 0, 0, 0, -1/16, -1/16, -1/16, -1/4, 7/16, 7/16, 7/16].$$

By the proof of [27, Theorem 3.18], the basic FPDims remain invariant under zesting. Furthermore, [27, Theorem 5.13] gives a zested modular fusion category whose topological spins match those in the third-to-last example of [3, §I.1.5]:

$$[0, 0, 0, 0, -5/16, 7/16, -5/16, -1/4, 3/16, -1/16, 3/16].$$

This result can be verified using the script `ZestD42.sage`, shared by César Galindo, located in the `Codes/SageMath` folder of the repository [49]. Note that these two modular fusion categories are not Galois conjugate, since the first has four distinct topological spins, while the second has six. The remaining six MDs are obtained through Galois conjugation. To see this, simply apply the following SageMath function to the lists above.

```
def GaloisConjugates(L):
    n=lcm([denominator(i) for i in L])
    R=[]
    for m in range(n):
        if gcd([m,n])==1:
            R.append([(n*m*i)%n/n for i in L])
    return R
```

Recall that the fusion data remains invariant under Galois conjugation; see [17] for further details.

4. EGYPTIAN FRACTIONS WITH SQUARED DENOMINATORS

A (q, r) -Egyptian fraction with squared denominators is a sum of the form

$$q = \sum_{i=1}^r \frac{1}{s_i^2},$$

where $q, r, s_i \in \mathbb{Z}_{\geq 1}$ and $s_1 \geq s_2 \geq \dots \geq s_r \geq 1$. In the context of classifying potential Grothendieck rings of modular integral fusion categories (or more generally, half-Frobenius integral fusion rings), we can assume that each s_i divides s_1 . By removing terms equal to 1, we can further reduce to the case $s_i \geq 2$ for all remaining terms, which immediately gives the bound $q \leq r/4$. The following proposition formalizes this reduction.

Proposition 4.1. *Consider a (q, r) -Egyptian fraction as defined above with $s_i \mid s_1$ for all i . Let k be the number of indices i with $s_i = 1$. Then the reduced $(q - k, r - k)$ -Egyptian fraction $q - k = \sum_{i=1}^{r-k} 1/s_i^2$ satisfies $s_i \geq 2$, inherits the divisibility condition, and obeys $q - k \leq (r - k)/4$. Conversely, any (q, r) -Egyptian fraction with these properties is obtained by appending k unit terms $1 = 1/1^2$ to a reduced $(q - k, r - k)$ -Egyptian fraction.*

Proof. Since $s_1 \geq \dots \geq s_r$, all terms equal to 1 occur at the end. Removing these k terms yields a $(q - k, r - k)$ -Egyptian fraction with $s_i \geq 2$ and $s_i \mid s_1$. For these remaining terms, $1/s_i^2 \leq 1/4$, so $q - k = \sum_{i=1}^{r-k} 1/s_i^2 \leq \sum_{i=1}^{r-k} 1/4 = (r - k)/4$. The converse is immediate: appending k unit terms to any reduced $(q - k, r - k)$ -Egyptian fraction recovers a valid (q, r) -Egyptian fraction satisfying the ordering and divisibility conditions. $\square$

The following steps outline our methodology for obtaining the 9025 possible types mentioned in §7.1:

- Employ the function `all_rep` (available in `rep_sq.sage` in [2]) for $1 \leq r \leq 13$ and $1 \leq q \leq r/4$.
- Refine the classification by incorporating additional 1s as described previously.
- Construct all possible types using $d_i = s_1/s_i$.

From rank 13 onward, we also used refined versions of the `all_rep` function, as explained below, all available also in `rep_sq.sage` in [2]. The module employs parallelization through SageMath's recursively enumerable sets (RES) functionality to enhance efficiency. The module provides the following general functions representing a given rational number as Egyptian fractions with squared denominators:

- `all_rep(q, N)`: computes all length- N representations of the rational number q .
- `all_cf_rep(q, cf)`: computes the representations of q with numerators specified by the list `cf`.

Additionally, the module offers specialized functions for computing representations that align with the fusion ring types discussed in the paper:

- `all_rep_MNSD(q, N)`: same as above under MNSD constraint, see §12.
- `all_rep_th87(N)`: computes the length- N representations satisfying Theorems 8.5 and 8.7.
- `all_rep_above_r(N)`: same as above where at least one denominator has prime factor $> N$, see §9.2.
- `all_rep_above_half_r(N)`: same as above where at least one denominator has prime factor $> N/2$.
- `all_rep_primes23(N)`: same as above where denominators are restricted to have prime factors 2 and 3 only.

Note that functions may include additional optional parameters not listed here, which can be utilized to refine the computations further.

5. TYPE CRITERIA

In this section, we delineate criteria that were employed to exclude certain candidates from being the type of a fusion ring. A *type* refers to a list denoted by $t = [[d_1, m_1], [d_2, m_2], \dots, [d_s, m_s]]$, where the conditions $1 = d_1 < d_2 < \dots < d_s$ and $m_i \geq 1$ for all indices i are satisfied. Such a type is characterized as:

- *trivial* if $t = [[1, 1]]$,
- *pointed* if $t = [[1, m]]$ for some m ,
- *perfect* if $m_1 = 1$,
- *integral* if each d_i is an integer.

A type $t = [[d_1, m_1], [d_2, m_2], \dots, [d_s, m_s]]$ may sometimes be represented simply as

$$[d_1, \dots, d_1, d_2, \dots, d_2, \dots, d_s, \dots, d_s],$$

where each d_i appears m_i times. Thus, we can rephrase the notation for a type of rank r as $[d'_1, \dots, d'_r]$ with the condition $1 = d'_1 \leq d'_2 \leq \dots \leq d'_r$. The criteria described in this section are primarily verified using modular arithmetic, with the sole exception of Proposition 5.2, and are presented in order of increasing computational complexity. These checks are implemented in the `TypeCriteria` function within the `TypeCriteria.sage` script, as referenced in [49]. Proposition 5.2 is valid at the level of fusion categories but does not hold for fusion rings. Accordingly, the function `TypeCriteriaRing` corresponds to `TypeCriteria` with this proposition omitted. Nevertheless, the proposition is expected to remain valid for half-Frobenius fusion rings; this has been verified for all cases up to rank 13 in [10]. Applying these criteria excludes 5655 of the 9025 types enumerated in §7.1, representing over 62% of the total, and completes in approximately one minute. The breakdown of excluded types is presented in Table 2. For the remaining types, we will utilize the fusion

Rank	1	2	3	4	5	6	7	8	9	10	11	12	13
# Types	1	1	1	1	2	3	3	7	11	42	144	812	7997
# Excluded Types	0	0	0	0	0	1	1	3	5	28	90	535	4992

TABLE 2. Number of excluded types per rank

ring solver, as elaborated in §6.

5.1. Small Perfect Type. In this subsection, we investigate perfect fusion rings under two constraints: either the number of distinct basic FPdims is small, or the global FPdim has few prime factors.

Theorem 5.1. *Any perfect integral fusion ring of type $[[d_1, m_1], [d_2, m_2], \dots, [d_s, m_s]]$, $s \leq 3$, is trivial.*

Proof. See the arXiv versions of this paper (up to v7) for details. □

Consequently, every non-trivial perfect integral fusion ring has rank at least 4, since a fusion ring of rank at most 3 has at most three distinct basic FPdims, and is therefore trivial by Theorem 5.1. This theorem cannot be extended to $s = 4$, as demonstrated by $\text{Rep}(A_5)$, the representation category of the alternating group A_5 , which has type $[[1, 1], [3, 2], [4, 1], [5, 1]]$. The proof of this theorem is deferred to the forthcoming paper [10], where it is more suitably placed. It effectively rules out only the following four types among the 9025 presented in §4,

$$[[1, 1], [2, 2], [3, 3]], \quad [[1, 1], [2, 6], [5, 3]], \quad [[1, 1], [2, 2], [3, 7]], \quad [[1, 1], [3, 7], [4, 5]].$$

All of these types are also excluded at the categorical level by Proposition 5.2, since their FPdim are of the form $p^a q^b$.

Proposition 5.2. *There exists no non-trivial perfect integral fusion category (over $\mathbb{C}$) with $\text{FPdim} = p^a q^b$.*

Proof. This follows directly from [24, Theorem 1.6 and Proposition 4.5(iv)]. □

We note that Proposition 5.2 does not generalize to fusion rings, as there exists a perfect fusion ring with $\text{FPdim} = 143 = 11 \cdot 13$ and type $[1, 4, 4, 5, 6, 7]$. This example is not 1-Frobenius. Whether the proposition extends to the 1-Frobenius case remains an open question. For reference, the extension holds in the half-Frobenius case up to rank 13 [10]. Theorem 5.1 and Proposition 5.2 serve as criteria checked by the function `SmallPerfect`:

```
sage: l=[1, 2, 2, 3, 3, 3]
sage: SmallPerfect(l)
False
```

5.2. Gcd Criterion.

Lemma 5.3. *Consider a non-pointed fusion ring of type $[d_1, d_2, \dots, d_r]$. For all i such that $d_i > 1$, let Z_i be the set of indices $j \neq 1$ for which N_{i,i^*}^j is nonzero, and let g_i be $\gcd_{j \in Z_i}(d_j)$. Then it holds that $d_i^2 \equiv 1 \pmod{g_i}$ and $\gcd(d_i, g_i) = 1$.*

Proof. First, note that Z_i is non-empty, which implies that $g_i \neq 0$. According to the Frobenius-Perron theorem, the dimension equation, and the Dual axiom, we have

$$d_i^2 = d_i d_{i^*} = \sum_k d_k N_{i,i^*}^k = 1 + \sum_{j \in Z_i} d_j N_{i,i^*}^j = 1 + K g_i,$$

where K is some integer. Consequently, $d_i^2 \equiv 1 \pmod{g_i}$, and $0 \equiv 1 \pmod{\gcd(d_i, g_i)}$. The lemma follows. $\square$

Proposition 5.4. *Consider a non-trivial perfect fusion ring of type $[d_1, d_2, \dots, d_r]$. Take $i > 1$, let Z'_i be the set of indices $j \neq 1$ for which $d_j < d_i^2$, and let g'_i be $\gcd_{j \in Z'_i}(d_j)$. Then $g'_i = 1$. In particular, $\gcd(d_2, \dots, d_r) = 1$.*

Proof. Note that if N_{i,i^*}^j is nonzero, then $d_i^2 \geq d_j$. Hence, following the notation in Lemma 5.3, Z_i is included in Z'_i , and as a result, g'_i divides g_i . Due to perfectness, we have $d_i > 1$, implying $d_i^2 > d_i$ and therefore i belongs to Z'_i . Consequently, g'_i divides d_i . However, according to Lemma 5.3, $g'_i = 1$. For the final assertion, note that $\gcd(d_2, \dots, d_r)$ is a divisor of $g'_2 = 1$. $\square$

The SageMath code implementing the criterion from Proposition 5.4 is located in the function `GcdCriterion` within the file `TypeCriteria.sage`, available at [49]. This criterion eliminates over 37% of the perfect types listed in §4, such as

```
sage: l=[1, 2, 2, 2, 6, 14, 14, 21, 21, 21]
sage: GcdCriterion(l)
False
```

The count per rank is presented in Table 3.

Rank	8	9	10	11	12	13
# Excluded Perfect Types	1	1	7	19	212	2474

TABLE 3. Number of excluded types per rank for the gcd criterion

5.3. Type Test. Let's consider a type $t = [d_1, \dots, d_r]$ with $1 = d_1 \leq \dots \leq d_r$ and $d_2 > 1$ (signifying that it is perfect). If there is an index i and $g_i > 1$ such that g_i divides every d_j not equal to 1 or d_i , and d_i is coprime with g_i , then assume a fusion ring of this type exists with a basis $\{b_1, \dots, b_r\}$ where $d_k = \text{FPdim}(b_k)$.

Lemma 5.5. *For every j with $d_j \neq 1$ and $d_j \neq d_i$, the following equation holds:*

$$\sum_{k; d_k = d_i} N_{j,j^*}^k \equiv -1/d_i \pmod{g_i}.$$

Proof. For each j with $d_j \neq 1$ and $d_j \neq d_i$, we have:

$$b_j b_{j^*} = b_1 + \sum_{k; d_k = d_i} N_{j,j^*}^k b_k + \sum_{k; d_k \neq 1, d_i} N_{j,j^*}^k b_k.$$

By applying FPdim and reducing modulo g_i , we obtain:

$$0 = 1 + x d_i \pmod{g_i},$$

where d_i has a multiplicative inverse modulo g_i . Therefore, $x \equiv -1/d_i \pmod{g_i}$. $\square$

Given an integer a_{d_i} such that $0 \leq a_{d_i} < g_i$ and $a_{d_i} \equiv -1/d_i \pmod{g_i}$, let S be the set containing all such d_i . From Lemma 5.5, for every $j \neq 1$, the inequality below must hold:

$$d_j^2 \geq 1 + \sum_{d \in S \setminus \{d_j\}} a_d d,$$

thus if the inequality does not hold, t cannot be a type of a fusion ring. Furthermore, if the set $\{k \mid d_k = d_j\}$ is a singleton, we can use a stronger inequality:

$$d_j^2 \geq 1 + b_j d_j + \sum_{d \in S \setminus \{d_j\}} a_d d,$$

with $0 \leq b_j < g_j^2$ and $b_j \equiv d_j - \frac{1}{d_j} \pmod{g_j^2}$.

The SageMath code implementing this criterion can be found in the function `TypeTest` within the file `TypeCriteria.sage`, available at [49]. This criterion eliminates over 36% of the perfect types listed in §4, such as

```
sage: T=[ [1,1], [5,1], [7,1], [35,1], [60,1], [140,2], [210,3] ]
sage: TypeTest(T)
False
```

The count per rank is presented in Table 4.

Rank	8	9	10	11	12	13
#Excluded Perfect Types	1	1	12	37	249	2380

TABLE 4. Number of excluded types per rank for the type test

5.4. Local Criterion. Consider a type $t = [[d_1, m_1], [d_2, m_2], \dots, [d_s, m_s]]$. Assume the existence of $g, i_0 > 1$ such that g divides each d_i for all indices i not in the set $\{1, i_0\}$, and d_{i_0} is coprime with g . Let $(d, m) := (d_{i_0}, m_{i_0})$. If it corresponds to a fusion ring with a basis $\{b_{1-m_1}, \dots, b_0, b_1, \dots, b_{r-1}\}$, where b_0 is the unit, $\text{FPdim}(b_i) = 1$ for $i \leq 0$, and $\text{FPdim}(b_j) = d$ for $j \in \{1, \dots, m\}$, then the following lemma applies:

Lemma 5.6. *For each $i \in \{1, \dots, m\}$, the equation below is valid:*

$$\sum_{j,k=1}^m N_{i,j}^k \equiv md - \frac{m_1}{d} \pmod{g^2},$$

and for all $j > m$, the integer g divides $\sum_{k=1}^m N_{i,j}^k$.

Proof. For any $i \in \{1, \dots, m\}$ and $j > m$, since $\text{FPdim}(b_i) \neq \text{FPdim}(b_j)$, by Corollary 2.9 and Frobenius reciprocity, we have:

$$b_i b_j = \sum_{k \geq 1} N_{i,j}^k b_k = \sum_{k=1}^m N_{i,j}^k b_k + \dots,$$

Applying FPdim and reducing modulo g , we conclude that:

$$d \sum_{k=1}^m N_{i,j}^k \equiv 0 \pmod{g},$$

which implies that g divides $\sum_{k=1}^m N_{i,j}^k$. For each $i \in \{1, \dots, m\}$, the sum over the basis elements yields:

$$b_{i^*} \sum_{k=1}^m b_k = \sum_{s \leq 0} b_s + \sum_{j=1}^m \left(\sum_{k=1}^m N_{i,j}^k \right) b_j + \sum_{j>m} \left(\sum_{k=1}^m N_{i,j}^k \right) b_j.$$

After applying FPdim , we obtain $md^2 = m_1 + xd + yg^2$, hence $x \equiv md - \frac{m_1}{d} \pmod{g^2}$. $\square$

For a type, we can analyze the partitions of $md^2 - xd - m_1$ in the form $\sum_{i \notin \{1, i_0\}} a_i d_i$, with $x \equiv md - \frac{1}{d} \pmod{g^2}$ and $a_i \equiv 0 \pmod{g}$. The SageMath code performing this analysis can be found in the function `LocalCriterion` within the file specified earlier, also available at [49]. This criterion, which can rule out types when no suitable partitions are found, is further detailed in the following example.

```
sage: T=[ [1,1], [1295,2], [3990,1], [4218,1], [24605,1], [42180,1], [98420,2], [147630,3] ]
```

We can apply Lemma 5.6 to the triples $(d, m, g) = (1295, 2, 19), (3990, 1, 37), (4218, 1, 5)$. Subsequently, we obtain $md - \frac{1}{d} \equiv 126, 1135, 11 \pmod{g^2}$ for each respective triple. The application of the function `LocalCriterion` to the triple $(d, m, g) = (1295, 2, 19)$ enables us to eliminate the type T in less than 0.1 second.

```
sage: %time LocalCriterion(T, 1295, 2, 19)
CPU times: user 82.8 ms, sys: 0 ns, total: 82.8 ms
Wall time: 82.3 ms
```

$\square$

However, we cannot employ the triple $(d, m, g) = (3990, 1, 37)$, as it yields 55 solutions.

```
sage: L = LocalCriterion(T, 3990, 1, 37); len(L)
```

55

The function `LocalCriterionAll` manages all operations. It requires only two inputs: a type and a time limit.

```
sage: %time LocalCriterionAll([[1, 1], [2, 2], [3, 3], [6, 4]],1)
CPU times: user 1.44 ms, sys: 4.4 ms, total: 5.84 ms
Wall time: 9.55 ms
False
```

Its application to the list in §4 led to the exclusion of several types per rank, as presented in Table 5. It is noteworthy

Rank	6	7	8	9	10	11	12	13
# Excluded Types	1	1	3	5	21	63	344	2852
# Excluded Perfect Types	1	1	2	2	14	37	238	2173

TABLE 5. Number of excluded types per rank for the local criterion

that this criterion alone suffices to eliminate all perfect types up to rank 9. Therefore, it can be stated conclusively that no non-trivial perfect integral half-Frobenius fusion rings, and thus no non-trivial perfect modular integral fusion categories, exist up to rank 9. The use of a fusion ring solver as detailed in §6 can extend this conclusion up to rank 12, as discussed in §7, and then to rank 13 in the categorical case, see §10.

6. ENHANCED FUSION RING SOLVER USING NORMALIZ

A fusion ring solver is a computational tool designed to receive a particular type as input and output all corresponding fusion rings of that type. Initially, §6.1 provides a brief introduction to the highly intuitive user interface of Normaliz [11] from version 3.10.2, and §6.2 offers an overview of Normaliz’s goals and outlines the adjustments made to support the linear and polynomial constraints specific to fusion rings. The last two subsections §6.3 and §6.4 contains the approach combining SageMath and Normaliz *before* version 3.10.2 (and show the system of equations explicitly). They introduce two versions of a fusion ring solver: the full version, which is discussed in §6.3 and addresses both dimension equations and associativity equations, and the partition (intermediate) version, which is detailed in §6.4 and focuses on a simplified set of dimension equations through the implementation of a partition.

6.1. Normaliz User Interface for Fusion Rings. Starting from version 3.10.2, Normaliz [11] offers a streamlined user interface for computing fusion rings. This is illustrated through the input file named `bracket_4.in`, found in the `example` directory of the Normaliz distribution:

```
amb_space auto
fusion_type
[1,1,2,3,3,6,6,8,8,12,12]
fusion_duality
[0,1,2,3,4,5,6,7,8,9,11,10]
```

It’s important to note that in the duality, indices start from 0. From this input, Normaliz generates the linear and quadratic equations defining the fusion data. The default computational goal for this input is `FusionRings`.

To run this command on Linux or MacOS, use the following command line syntax, assuming `example` is the current directory. For some progress information on the terminal, you can add the `-c` flag. On a modern laptop, the computation typically takes less than 10 seconds and requires about 2.4 GB of RAM.

```
path/to/normaliz bracket_4
```

A brief informal explanation of the algorithm used to solve the system of equations is presented in §6.2.

The results are detailed in the `bracket_4.out` file, beginning with a preamble:

```
148 fusion rings up to isomorphism
0 simple fusion rings up to isomorphism
148 nonsimple fusion rings up to isomorphism
Embedding dimension 231
dehomogenization
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ... 0 0 0 0 0 1
```

The 148 fusion rings correspond to the orbits of the set of lattice points with respect to the symmetries of the equation system. These symmetries are observed under permutations of the type vector that adhere to the Frobenius-Perron equations and are compatible with the duality. Put simply, we have identified 148 pairwise nonisomorphic fusion rings, classified according to their type and duality. They are automatically categorized into simple and nonsimple fusion rings. To limit the computation exclusively to simple fusion rings, one can modify the input file by including the `SimpleFusionRings` option.

The term *embedding dimension* refers to the number of coordinates utilized during the computation. The rationale behind the selection of these coordinates is discussed in §6.3. The final component, represented by the number 1 in

the dehomogenization process, signifies that the equations' right-hand side corresponds to the last coordinate of the solutions. This component is not included in the fusion data.

The latter part of the output file details the fusion rings represented by lattice points:

```
0 simple fusion rings up to isomorphism:

148 nonsimple fusion rings up to isomorphism:
0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 ... 1 3 3 1 1 1
...
```

To generate fusion data from the lattice points, add the `FusionData` option in the input file. The result includes a list of fusion data for each fusion ring, presented as a series of matrices M_i , where $i = 1, \dots, r$ and r is the rank of the fusion ring ($r = 12$ in our example). The matrix M_i comprises the elements N_{ij}^k , with row index k and column index j .

The input file for solving the dimension partition version (see §6.4), for instance, `bracket_3_part.in`, is as follows:

```
amb_space auto
fusion_type_for_partition
[1,1,2,3,3,6,6,8,8,8,12,12]
```

Here, the default computation goal is `SingleLatticePoint`, focusing on the solvability of the system.

For additional information and further options, refer to Appendix H of the Normaliz manual ([Normaliz.pdf](#)), available in the `doc` directory of the Normaliz distribution or online at [12]. This includes details on restricting computations to fusion rings that meet certain criteria for modular categorification.

6.2. Normaliz and its Approach to Fusion Rings. Normaliz [11] is an open source software for discrete convex geometry and its algebraic aspects. Readers are referred to Bruns and Gubeladze [9] for detailed terminology and a comprehensive discussion. Normaliz is designed to solve Diophantine systems of linear inequalities, equations, and congruences with integer coefficients. Additionally, it calculates enumerative information such as multiplicities (which correspond to geometric volumes) and Hilbert series. Objects in Normaliz can be defined either by generators, such as the extreme rays of cones, bases of lattices, and vertices of polytopes, or by constraints like inequalities, equations, and congruences. For systems with coefficients in real algebraic number fields, Normaliz can execute fundamental operations like convex hull computation and its dual, vertex enumeration. Moreover, it is capable of computing lattice points within (bounded) polytopes over real algebraic number fields, facilitating applications to non-integral fusion rings. In the context of fusion rings, it is crucial that lattice points within polytopes can be subjected to constraints imposed by polynomial equations and inequalities. Each release of Normaliz includes source code, comprehensive documentation, sample examples, a testing suite, and pre-compiled binaries for Linux, Mac OS, and MS Windows systems.

For lattice points in generic polytopes denoted by P , Normaliz employs the project-and-lift algorithm. It sequentially projects P onto coordinate hyperplanes until reaching zero dimensions and then lifts the lattice points back up. If P' is a projection of P onto a coordinate hyperplane, then the lattice points of P are projected to lattice points in P' , and if $x \in P'$ is a lattice point within P' , its preimages are the lattice points in a line segment. Polynomial constraints can be introduced as soon as the lifting process reaches the highest coordinate present in the constraint.

In its standard form, the project-and-lift method is suitable for only minor cases of fusion rings. For satisfactory performance, the algorithm has been tailored to the special linear and polynomial constraint structure specific to fusion rings. Each linear equation is inhomogeneous with nonnegative coefficients and a positive right-hand side. We can refer to the set of coordinates that appear in the equation with positive coefficients as a "patch". These patches encompass the entire set of coordinates, and thus the linear equations, when restricted to the nonnegative orthant, delineate a polytope P . Solutions to a linear equation, confined to its patch, are ascertained using the project-and-lift technique previously outlined, and the lattice points in P are derived by combining these local solutions along matching components. In essence, we begin with the solutions of one of the equations and progressively extend them patch by patch. The sequence in which patches are integrated into the extension process is pivotal. Normaliz includes options that allow alteration of the sequence, as detailed later on.

In the partition version (§6.4), the input file is only made of simplified dimension (linear) equations. Particularly for this case, it is critical to recognize a secondary, implicit constraint type: congruences extracted from the linear equations by taking successive residue classes modulo their coefficients. By default, each congruence involves only the coordinates pertaining to the patch of its originating equation. Nonetheless, since congruences only involve a subset of these coordinates, they frequently pertain to other patches or combinations thereof, potentially significantly limiting their number of solutions. Our current classification up to rank 13 would not have been achievable without meticulous utilization of the congruences.

When polynomial equations of degree two or higher are in play, Normaliz endeavors to determine an optimal patch extension order that allows these equations to be applied as early as feasible. Users can influence this order by either insisting on the "linear" input order or by directing Normaliz to employ "weights" that gauge the anticipated solution count for each patch and prioritize those with lower weight. Regardless of whether polynomial equations are present, users can request an order based on the applicability of congruences. This order can also be weight-dependent.

Some computations were executed on the high-performance cluster (HPC) at Osnabrück by early splitting of partial solutions into parts, which were then processed separately. Despite the rather basic approach of using a static subdivision without intercommunication between running instances of Normaliz, the HPC proved to be advantageous.

6.3. Full Version. *All the processes outlined in this subsection have been fully automated in Normaliz [11] from version 3.10.2, as outlined in §6.1. Appendix H of its manual [12] specifically addresses the computation of the fusion rings for a specified type.*

Consider a fusion ring with the basis $\{b_1, \dots, b_r\}$. As described in §2.1, for all indices i, j :

$$b_i b_j = \sum_k N_{i,j}^k b_k,$$

and by applying `FPdim`, we obtain the type $[d_1, \dots, d_r]$ and the corresponding *dimension equations*:

$$d_i d_j = \sum_k N_{i,j}^k d_k.$$

The objective is to resolve these r^2 linear positive Diophantine equations, where (d_i) are specified and $(N_{i,j}^k)$ represent r^3 variables, using Normaliz. Now, we can decrease the variable count to roughly $(r-1)^3/6$ by invoking the Unit axiom ($N_{1,i}^j = N_{i,1}^j = \delta_{i,j}$) from the Definition 2.1 of fusion data, as well as the Frobenius reciprocity (Proposition 2.2).

A critical factor in accelerating computation is the strategic use of associativity equations (non-linear)

$$\sum_s N_{i,j}^s N_{s,k}^t = \sum_s N_{j,k}^s N_{i,s}^t,$$

in the most effective manner possible during the solving process of the aforementioned linear Diophantine equations. While the optimal approach is not confirmed, the *patching* method we employ is highly efficient (refer to §6.2 for further details).

In practice, for a given type $L = [d_1, d_2, \dots, d_r]$, utilize the function `TypeToNormaliz`, the SageMath code for which can be found at [49]. This function generates input files (.in), one for each potential duality map $i \rightarrow i^*$. Place these files in a directory alongside the `normaliz.exe` and `run_normaliz.bat` files available at [49], and execute `run_normaliz` (note the existence of a more recent and faster Linux version used for our latest computations). This process yields output files (.out) containing all potential solutions (if any exist). The remaining task is to convert these solutions into fusion data, considering isomorphism. We demonstrate how this can be done with the following example. Take the type $L = [1, 1, 2]$ of the character ring of S_3 . When `TypeToNormaliz` is applied, it generates the file `[1,1,2][0,1,2].in` with the content as follows:

```
amb_space 4
inhom_equations 4
1 2 0 0 0
0 1 2 0 -2
0 1 2 0 -2
0 0 1 2 -3
LatticePoints
convert_equations
nonnegative
polynomial_equations 2
x[2]^2 - x[1]*x[3] + x[3]^2 - x[2]*x[4] - 1;
-x[2]^2 + x[1]*x[3] - x[3]^2 + x[2]*x[4] + 1;
```

The upper part encodes the linear Diophantine equations, and the lower part lists the associativity equations. Following the execution of `run_normaliz`, the file `[1,1,2][0,1,2].out` is produced, containing:

```
1 lattice points in polytope (module generators) satisfying polynomial constraints:
0 0 1 1 1
```

Here, we encounter a single solution, but there could be multiple in general (as seen in the subsequent example). Next, remove the final '1' from each line of the solution and convert it into a list of lists:

```
sage: LL=[[0,0,1,1]]
```

Collect the lists for the type and the duality map:

```
sage: L=[1,1,2]
```

```
sage: d=[0,1,2]
```

Finally, to obtain all the fusion data up to isomorphism, apply the function `ListToFusion`:

```
sage: ListToFusion(LL,L,d)
[[[1, 0, 0], [0, 1, 0], [0, 0, 1]],
 [[0, 1, 0], [1, 0, 0], [0, 0, 1]],
 [[0, 0, 1], [0, 0, 1], [1, 1, 1]]]
```

$$\begin{array}{ccccccccc} 1 & 0 & 0 & & 0 & 1 & 0 & & 0 & 0 & 1 \\ 0 & 1 & 0 & , & 1 & 0 & 0 & , & 0 & 0 & 1 \\ 0 & 0 & 1 & & 0 & 0 & 1 & & 1 & 1 & 1 \end{array}$$

6 lattice points in polytope (module generators) satisfying polynomial constraints:

[illegible][illegible]
$$T = [[1, m_1], [d_2, m_2], \dots, [d_s, m_s]].$$

We can reframe the type as $[1, d_{1,1}, \dots, d_{1,n_1}, d_{2,1}, \dots, d_{2,n_2}, \dots, d_{s,1}, d_{s,n_s}]$, where $d_{i,a} = d_i$, $d_1 = 1 = d_{0,1}$, and $n_i = m_i - \delta_{1,i}$. The dimension equations are then expressed as follows:

$$d_{i,a}d_{j,b} = \sum_{k,c} N_{i,a,j,b}^{k,c} d_{k,c}.$$

Let us define $D_i := \sum_{a=1}^{n_i} d_{i,a} = n_i d_i$ and $M_{i,j}^k := \sum_{a,b,c} N_{i,a,j,b}^{k,c}$, which simplifies the equations to:

$$D_i D_j = \sum_{a,b} \sum_{k,c} N_{i,a,j,b}^{k,c} d_{k,c} = \sum_k (\sum_{a,b,c} N_{i,a,j,b}^{k,c}) d_k = \sum_k M_{i,j}^k d_k.$$

$$n_i d_i n_j d_j = \sum_k M_{i,j}^k d_k,$$

Lemma 6.1. *The following equalities hold:*

- (*Unit*) $M_{i,0}^j = M_{0,i}^j = \delta_{i,j} m_i$
- (*Dual*) $M_{i,j}^0 = M_{j,i}^0 = \delta_{i,j} m_i$
- (*Frobenius reciprocity*) $M_{i,j}^k = M_{i,k}^j = M_{j,k}^i = M_{j,i}^k = M_{k,i}^j = M_{k,j}^i.$

☐

Remark 6.2. While this version utilizes the dimension partition of the type, alternative versions could explore other pertinent partitions.

7. HALF-FROBENIUS INTEGRAL FUSION RINGS UP TO RANK 12

This section focuses on classifying all half-Frobenius integral fusion rings up to rank 12. Initially, we considered 1028 types derived from Egyptian fractions with squared denominators, as discussed in §7.1. From these, we identified 71 types of fusion rings, but none were perfect, thus proving Theorem 1.6. Subsequently, we classified 10628 fusion rings originating from these 71 types, detailed in §7.2. Among them, we identified 213 noncommutative fusion rings. Ultimately, only 69 fusion rings, from 27 types, are commutative, cyclotomic, and self-transposable, as outlined in §7.3.

7.1. List of Possible Types. Based solely on Egyptian fractions with squared denominators, we found 1028 possible types up to rank 12 (in fact, 9025 ones up to rank 13). The counting per rank is presented in Table 6.

Rank	1	2	3	4	5	6	7	8	9	10	11	12	13
#Types	1	1	1	1	2	3	3	7	11	42	144	812	7997
#Perfect Types	1	0	0	0	0	1	1	2	2	24	88	591	6517

TABLE 6. Number of possible types per rank

The comprehensive list of possible types up to rank 13, available in `ListPossibleTypes.tar.xz` in [49], was computed using `all_rep` from §4. The ratio of perfect types (refer to §5) exhibits an increasing trend, e.g. 18% for rank 9, but 81% for rank 13. This leads us to question whether this ratio tends to 1 as the rank goes to infinity.

7.2. List of Fusion Rings. All computational steps are documented in `InvestHFuptoR12.txt`, as referenced in [49]. The initial step involved processing the 1016 non-pointed types up to rank 12 mentioned in §7.1. From these, merely 352 types met the `TypeCriteria` outlined in §5. Utilizing the partition version of our fusion ring solver, as discussed in §6.4 and limiting the processing time to ten seconds per type, further reduced the count to just 77 types. Subsequently, excluding the type $[1, 1, 63, 135, 140, 252, 540, 1260, 1260, 1890, 1890, 1890]$ with HPC brought the total down to 76 types. The distribution of these types across ranks 5 to 12 is 1, 1, 1, 3, 5, 9, 17, 39, respectively. Out of these, only $[1, 2, 2, 3, 4, 4, 15, 15, 20, 30, 30, 30]$ is perfect. Following this, we employed the full version of our fusion ring solver, as outlined in §6.3, which necessitated specifying duality. This adjustment resulted in the expansion of our 76 types into 801 cases. We streamlined these cases using the `SingleLatticePoint` option, as elucidated in §6.1. After an initial round of processing, limited to 10 seconds per case, 50 cases remained unresolved, plus 312 cases with a solution. The only perfect type is already excluded at this step, which proves Theorem 1.6. Further rounds of processing, with limits set to 100 seconds and then 1000 seconds, resulted in 10 then 3 unresolved cases, respectively. Ultimately, we solved the last cases with HPC, concluding with solutions for 340 cases across 59 distinct non-pointed types, as listed below for each rank, arranged in lexicographic order:

- Rank 5: $[[1, 1, 1, 1, 2]]$,
- Rank 6: $[[1, 1, 1, 1, 2, 2]]$,
- Rank 7: $[[1, 1, 1, 1, 2, 2, 2]]$,
- Rank 8: $[[1, 1, 1, 1, 2, 2, 2, 2], [1, 1, 1, 1, 2, 2, 2, 4], [1, 1, 2, 2, 2, 2, 3, 3]]$,
- Rank 9: $[[1, 1, 1, 1, 2, 2, 2, 2, 2], [1, 1, 1, 1, 2, 2, 2, 4, 4], [1, 1, 1, 1, 4, 4, 6, 6], [1, 1, 2, 2, 2, 2, 3, 3, 6]]$,
- Rank 10: $[[1, 1, 1, 1, 1, 1, 1, 1, 1, 3], [1, 1, 1, 1, 1, 1, 1, 1, 2, 2], [1, 1, 1, 1, 2, 2, 2, 2, 2, 2], [1, 1, 1, 1, 2, 2, 2, 4, 4], [1, 1, 1, 1, 4, 4, 6, 6, 12], [1, 1, 1, 2, 2, 2, 2, 2, 3], [1, 1, 2, 2, 2, 2, 2, 3, 3, 6, 6], [1, 1, 2, 3, 3, 4, 4, 4, 6, 6]]$,
- Rank 11: $[[1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 3], [1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2], [1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 4], [1, 1, 1, 1, 1, 1, 2, 2, 2, 3, 3], [1, 1, 1, 2, 2, 2, 2, 2, 2, 3, 6], [1, 1, 1, 1, 2, 2, 2, 4, 4, 4, 4], [1, 1, 1, 1, 2, 2, 2, 4, 4, 4, 8], [1, 1, 1, 1, 2, 4, 4, 4, 4, 6, 6], [1, 1, 2, 2, 2, 2, 3, 3, 6, 6, 6], [1, 1, 2, 3, 3, 4, 4, 4, 6, 6, 12], [1, 1, 1, 1, 2, 6, 6, 8, 12, 12, 12], [1, 1, 1, 1, 4, 4, 6, 6, 6, 12, 12], [1, 1, 1, 3, 4, 4, 4, 4, 4, 6, 6], [1, 1, 1, 1, 4, 4, 12, 12, 18, 18, 18]]$,
- Rank 12: $[[1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 3], [1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2], [1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 4, 4], [1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2], [1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 4], [1, 1, 1, 1, 2, 2, 2, 2, 4, 6, 6, 6], [1, 1, 1, 1, 2, 2, 2, 4, 4, 4, 4, 4], [1, 1, 1, 1, 2, 2, 2, 4, 4, 4, 8, 8], [1, 1, 1, 1, 2, 2, 2, 8, 8, 12, 12, 12], [1, 1, 1, 1, 2, 4, 4, 4, 4, 6, 6, 12], [1, 1, 1, 1, 2, 6, 6, 8, 12, 12, 12, 24], [1, 1, 1, 1, 2, 8, 18, 18, 24, 36, 36, 36], [1, 1, 1, 1, 3, 3, 3, 3, 4, 4, 6, 6], [1, 1, 1, 1, 4, 4, 6, 6, 6, 12, 12, 12], [1, 1, 1, 1, 4, 4, 12, 12, 18, 18, 18, 36], [1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 6, 6], [1, 1, 1, 1, 2, 2, 2, 3, 4, 4, 4, 6, 6], [1, 1, 1, 3, 4, 4, 4, 4, 4, 6, 12], [1, 1, 1, 3, 6, 8, 8, 8, 8, 8, 12], [1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3], [1, 1, 2, 2, 2, 2, 3, 3, 6, 6, 6, 6], [1, 1, 2, 2, 2, 2, 3, 3, 6, 6, 6, 12], [1, 1, 2, 2, 2, 2, 6, 6, 6, 6, 9, 9], [1, 1, 2, 3, 3, 4, 4, 4, 6, 6, 12, 12], [1, 1, 2, 3, 3, 6, 6, 8, 8, 8, 12, 12], [1, 1, 2, 6, 6, 6, 6, 10, 10, 10, 15, 15]]$.

As previously mentioned, none are perfect. It is noteworthy that the perfect integral modular fusion category, and therefore half-Frobenius, $\mathcal{Z}(\text{Rep}(A_5))$ has an FPdim of $60^2 = 3600$, a rank of 22, and a type of $[[1, 1], [3, 2], [4, 1], [5, 1], [12, 10], [15, 4], [20, 3]]$, calculated using [36, Section 3] and GAP.

Question 7.1. *Is there a perfect integral half-Frobenius fusion ring/category with a rank less than 22?*

Note that the perfect integral modular fusion category $\mathcal{Z}(\text{Rep}(A_7))$, with FPdim $(7!/2)^2$, rank 74, and type:

$$[[1, 1], [6, 1], [10, 2], [14, 2], [15, 1], [21, 1], [35, 1], [70, 9], [105, 4], [210, 20], [280, 9], [360, 14], [504, 5], [630, 4]],$$

notably lacks any prime-power basic FPdim.

Question 7.2. *Is there a perfect integral half-Frobenius fusion ring/category, without any prime-power basic FPdim, that has a rank lower than 74?*

We ended with 71 types (pointed included), and the computation with the `FusionData` option (retricted to the 340 cases mentioned above, in the non-pointed case) provides 10628 fusion rings, 213 among them being noncommutative. The counting per rank is presented in Table 7. The list of half-Frobenius integral fusion rings up to rank 12 is available

Rank	1	2	3	4	5	6	7	8	9	10	11	12
#Types	1	1	1	1	2	2	2	4	5	9	15	28
#Fusion Rings	1	1	1	2	3	6	9	23	105	158	1218	9101
#Noncommutative	0	0	0	0	0	1	0	4	5	7	38	158

TABLE 7. Number of half-Frobenius fusion rings per rank

in `HalfFrobIntUpToRank12.tar.xz` in [49].

7.3. Commutative, Cyclotomic and Self-Transposable. Among the 10628 half-Frobenius integral fusion rings up to rank 12 discovered above, only 99.3% are commutative cyclotomic and self-transposable (see §3.1), i.e., 69 fusion rings from 27 types. The counting per rank is presented in Table 8. The types mentioned above, restricted to the

Rank	1	2	3	4	5	6	7	8	9	10	11	12
#Types	1	1	1	1	1	1	2	2	2	4	5	6
#Fusion Rings	1	1	1	2	1	1	3	7	4	11	13	24

TABLE 8. Number of commutative, cyclotomic and self-transposable half-Frobenius fusion rings per rank

non-pointed ones, are listed below:

- Rank 7: [1, 1, 1, 1, 2, 2, 2],
- Rank 8: [1, 1, 2, 2, 2, 2, 3, 3],
- Rank 9: [1, 1, 1, 1, 4, 4, 6, 6, 6],
- Rank 10: [1, 1, 1, 1, 2, 2, 2, 4, 4, 4], [1, 1, 1, 2, 2, 2, 2, 2, 2, 3], [1, 1, 2, 3, 3, 4, 4, 4, 6, 6],
- Rank 11: [1, 1, 1, 1, 2, 2, 2, 2, 2, 2], [1, 1, 1, 1, 2, 6, 6, 8, 12, 12, 12], [1, 1, 1, 3, 4, 4, 4, 4, 4, 6], [1, 1, 1, 1, 4, 4, 12, 12, 18, 18, 18],
- Rank 12: [1, 1, 1, 1, 2, 8, 18, 18, 24, 36, 36, 36], [1, 1, 1, 3, 6, 8, 8, 8, 8, 8, 12], [1, 1, 2, 2, 2, 2, 6, 6, 6, 6, 9, 9], [1, 1, 2, 3, 3, 6, 6, 8, 8, 8, 12, 12], [1, 1, 2, 6, 6, 6, 6, 10, 10, 10, 15, 15].

The full list of non-pointed commutative, cyclotomic, self-transposable half-Frobenius integral fusion rings up to rank 12 is available in `NonPtCycloHalfFrobSelfTransUpToRk12.tar.xz` in [49]. They were classified utilizing the list from §7.2 in conjunction with the functions `NonCo`, `ExtendedCyclo`, `ModularCriterion` and `SelfTransposable` contained within the file `ModularData.sage`, also available at [49].

8. ADVANCED RESULTS ON MODULAR FUSION CATEGORIES

From rank 13 onwards, it becomes impractical to classify all half-Frobenius integral fusion rings using our current technology. Thus, the types were further restricted by additional properties coming from more advanced results on modular fusion categories, which involved the universal grading §8.1, congruence representations of the modular group §8.2 and Galois action §8.3.

8.1. Universal Grading. Let G be a finite group. A G -grading of a fusion ring R is given by a partiton of its basis $B = \sqcup_{g \in G} B_g$ such that:

- For any $x \in B_g$ and any $y \in B_{g'}$, the basic components of xy belong to $B_{gg'}$.
- For any $x \in B_g$, x^* is in $B_{g^{-1}}$.

A G -grading is called *faithful* if B_g is non-empty for all $g \in G$. Consequently, by [22, Theorem 3.5.2], $\text{FPdim}(B_g) := \sum_{x \in B_g} \text{FPdim}(x)^2$ is constant in g . The faithful grading with the largest group is called the *universal grading*. By [22, Lemma 8.22.9], the universal grading group of the Grothendieck ring of modular fusion category is $G = B_{pt}$, the group of basic element with $\text{FPdim} = 1$ (see Corollary 2.10). All these modular constraints lead to the following definition:

Definition 8.1. Let t be a *type*, i.e., a sorted list of integers starting with 1. Let r be the length of t . Let p be the number of entries equal to 1 in t . Let $D := \sum_{d \in t} d^2$. A *modular partition* of t is a list L of lists such that:

- t is the sorted concatenation of the lists in L ,

- L is lexicographically sorted,
- the lists in L are sorted,
- L has p elements,
- p divides D ,
- for all l in L , then $\sum_{d \in l} d^2 = D/p$,

A solution for t may be called a *partitioned type*.

The function classifying all the modular partitions of a given type is named `ModularPartitions` in `TypeCriteria.sage` in [49]. Here are a few examples with 0, 1, 2, and 3 solutions:

```
sage: %attach TypeCriteria.sage
sage: L0=[1,1,1,1,2]
sage: ModularPartitions(L0)
[]
sage: L1=[1, 1, 1, 3, 12, 12, 30, 40, 40, 40, 40, 40, 60]
sage: ModularPartitions(L1)
[[[1, 1, 1, 3, 12, 12, 30, 60], [40, 40, 40], [40, 40, 40]]]
sage: L2=[1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 4, 4, 4]
sage: ModularPartitions(L2)
[[[1, 1, 1, 1, 4], [2, 2, 2, 2, 2], [2, 4], [2, 4]],
 [[1, 1, 1, 1, 2, 2, 2, 2], [2, 4], [2, 4], [2, 4]]]
sage: L3=[1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 6, 6]
sage: ModularPartitions(L3)
[[[1, 1, 2, 2, 2, 2, 3, 3, 6], [3, 3, 3, 3, 6]],
 [[1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3], [6, 6]],
 [[1, 2, 2, 3, 3, 3, 6], [1, 2, 2, 3, 3, 3, 6]]]
```

The classification of fusion rings constrained to such a grading, with grading group of order less than 5, is a new Normaliz feature, see [12, §H.5.2]. The following theorem is essentially a reformulation of [38, Proposition VI.2].

Theorem 8.2. *Let $\mathcal{C}$ be an integral modular fusion category. Let R be its Grothendieck ring with basis B . Let $G = B_{pt}$ be the universal grading group. Let $t_g := (\text{FPdim}(x))_{x \in B_g}$. Let $\mathcal{C}_e$ be the fusion subcategory corresponding to B_e . Then:*

- (0) *If $\mathcal{C}_e$ is perfect then it is modular,*
- (1) *If $B_{pt} \subset B_e$ and t_e has an entry with odd multiplicity, then $\forall g \neq e$, every entry of t_g has multiplicity ≥ 2 ,*
- (2) *If condition (1) is satisfied, $p := |B_{pt}|$ is prime, and an entry d in t_e appears with multiplicity m , then p divides d or m ,*
- (3) *If condition (1) is satisfied, $p := |B_{pt}|$ is prime, then there exists a modular fusion category $\mathcal{D}$ with $\text{FPdim} = \text{FPdim}(\mathcal{C})/p^2$ and type t'_e , where t'_e is a reduction of t_e that involves mapping p identical entries $x, \dots, x$ to a single entry x , or alternatively, mapping one entry x which is a multiple of p to p entries $x/p, \dots, x/p$.*

Remark 8.3. Since the Grothendieck ring of a modular fusion category $\mathcal{C}$ is half-Frobenius, for each entry x of its type, x^2 divides $\text{FPdim}(\mathcal{C})$. However, regarding Theorem 8.2 (3), it is often the case that an entry x in t_e does not satisfy the condition that x^2 divides $\text{FPdim}(t'_e) = \text{FPdim}(t_e)/p$, whereas t'_e must be half-Frobenius, therefore, this entry must be split into p entries $x/p, \dots, x/p$ in t'_e . This helps to reduce the number of possible t'_e .

The function `GradingCriteria` in `TypeCriteria.sage` automates the use of Theorem 8.2, it also iterates over possible types of modularization and rechecks `Theorem8_7Check` (from Theorem 8.7) and `TypeCriteria`.

Here are four examples illustrating the exclusion criteria, each corresponding to a point of Theorem 8.2.

- (0) In the partitioned type $[[1, 2, 2, 3, 3, 3, 3], [1, 2, 2, 3, 6]]$, the neutral component is perfect, but we already know that there is no perfect integral modular fusion category of rank 8,
- (1) In $[[1, 1, 1, 1, 2, 2, 2, 3, 10, 10], [15], [15], [15]]$, the pointed part is in the neutral component t_e , and the entry 2 has multiplicity three (odd) in t_e , but 15 appears with multiplicity one in some non-neutral components.
- (2) In $[[1, 1, 2, 2, 3, 3, 5, 6, 6, 10, 15], [15], [15]]$, the pointed part is in the neutral component t_e , and the entry 5 has multiplicity one (odd) in t_e , but 5 is not divisible by the prime $2 = |B_{pt}|$.
- (3) In $[[1, 1, 2, 12, 15, 15, 20, 20, 20], [30, 30]]$, the pointed part is in the neutral component t_e , and the entry 2 has multiplicity one (odd) in t_e . Now, $\text{FPdim}(t_e)/2 = 2^2 3^2 5^2$ which is not a multiple of 20^2 , so the entries 20 must split for the modularization type $t'_e = [1, 1, 1, 6, 6, 10, 10, 10, 10, 10, 15]$. However, t'_e is excluded by `TypeCriteria`.

8.2. Congruence Representation. This subsection reviews some applications of congruence representations of the modular group to modular fusion categories, leading to a proof of the folklore Theorem 8.4. Although a more concise proof is presented later in §8.3, the current exposition is meant to be informative and to serve for future research.

As discussed in [37, Section 3], a modular fusion category $\mathcal{C}$ is associated with modular data (S, T) , which gives a projective representation of

$$\mathrm{SL}(2, \mathbb{Z}) = \langle s, t \mid (st)^3 = s^2, s^4 = e \rangle.$$

This representation can be lifted to a usual (linear) representation ρ by utilizing the linear characters (i.e. one-dimensional representations), forming a cyclic group of order 12. This representation is r -dimensional—where r represents the rank of $\mathcal{C}$ —and is *congruence*. This means it factors through $\mathrm{SL}(2, \mathbb{Z}/n\mathbb{Z})$, for some n whose smallest one is called the *level*. The level is determined as $\mathrm{ord}(\rho(t))$ and satisfies

$$\mathrm{ord}(T) \mid \mathrm{ord}(\rho(t)) \mid 12\mathrm{ord}(T).$$

A finite-dimensional congruence representation ρ of level n is completely reducible, hence it can be broken down into a direct sum of irreducible representations of $\mathrm{SL}(2, \mathbb{Z}/n\mathbb{Z})$. It's important to note that this includes only those irreducible representations that do not further factor through $\mathrm{SL}(2, \mathbb{Z}/d\mathbb{Z})$ for any proper divisor d of n . Nevertheless, if $n = \prod_i p_i^{n_i}$ represents the prime factorization of n , then $\rho = \bigotimes_i \rho_i$ with each ρ_i being a congruence representation of level $p_i^{n_i}$.

For deeper applications, note that [39] proves that the finite-dimensional congruence representations are equivalent to *symmetric* ones, which are classified in [40].

The dimensions d of the irreducible finite-dimensional congruence representations at level $n = p^a$ are listed in the table at the end of [44]. For $a = 1$, we have $d \geq (n - 1)/2$, which implies $p = n \leq 2d + 1$. For $a \geq 2$, we have

$$d \geq \frac{n}{2} \left(1 - \frac{1}{p^2}\right),$$

which leads to

$$p^a = n \leq 2d + \frac{2d}{p^2 - 1}.$$

If $p > 2d + 1$, then $2d < p - 1$, which implies

$$p < p^a < 2d + \frac{1}{p+1} < 2d + 1,$$

a contradiction. Therefore, $p \leq 2d + 1$ for all a . Since the rank r of the modular fusion category is the sum of the dimensions d of these irreducible representations, we conclude that $d \leq r$, and thus $p \leq 2r + 1$.

According to Cauchy's theorem in [7], the set S of prime factors of $\mathrm{ord}(T)$ coincides with the set of prime factors of the global dimension norm N of the modular fusion category of rank r . The prime numbers p mentioned earlier (satisfying $p \leq 2r + 1$) form the set S' , which consists of the prime factors of the level n of the congruence representation.

Since $\mathrm{ord}(T) \mid n \mid 12\mathrm{ord}(T)$ and $12 = 2^2 \cdot 3$, it follows that $S \subseteq S' \subseteq S \cup \{2, 3\}$. Therefore, for all prime factors $p \neq 2, 3$ of N , we have $p \leq 2r + 1$. This inequality holds trivially for $p = 2, 3$.

This concludes our first proof of Theorem 8.4 without relying on [21, Theorem II (iii)].

8.3. Galois Action. Let (s, t) denote a *normalized* modular data, and let $\mathbb{Q}_n$ be the cyclotomic field $\mathbb{Q}(\zeta_n)$, where $n = \mathrm{ord}(t)$. In this subsection, σ will refer to a Galois automorphism in $\mathrm{Gal}(\mathbb{Q}_n/\mathbb{Q})$. For simplicity, we use the same symbol σ to denote the induced permutation $X \mapsto \sigma(X)$ on the simple objects. It acts on $\dim$, s , and t as follows:

- (1) $\sigma(\dim(X)^2) = \frac{\sigma(\dim(\mathcal{C}))}{\dim(\mathcal{C})} \dim(\sigma(X))^2$, see [22],
- (2) $\sigma(s_{X,Y}^2) = s_{X,\sigma(Y)}^2$, see [22],
- (3) $\sigma^2(t_X) = t_{\sigma(X)}$, see [21, Theorem II (iii)].

See for example [50, Section 2] for an explicit normalization of the modular data. The following result is a straightforward consequence of (3) and [7].

Theorem 8.4. *For any prime factor p of the dimension norm of a modular fusion category with rank r , it holds that $p \leq 2r + 1$.*

Proof. If $p = 2, 3$ then $p \leq 2r + 1$ trivially as $r \geq 1$. Let $p \neq 2, 3$ be a prime factor of the global dimension norm. By Cauchy theorem in [7], p divides $\mathrm{ord}(t)$. So there must be a simple object X such that p divides the order of t_X , thus the orbit of $(\sigma^2(t_X))$ has at least $(p - 1)/2$ distinct elements, because the group of units in $\mathbb{Z}/p\mathbb{Z}$ is cyclic of order $p - 1$, so it has an element g with $\mathrm{ord}(g^2) = (p - 1)/2$. So by (3), $r \geq (p - 1)/2$, i.e., $p \leq 2r + 1$. $\square$

Here is an example of type of rank $r = 13$ and $\mathrm{FPdim} = 2^4 3^2 5^2 7^2 19^2 37^2$ excluded Theorem 8.4:

$$[1, 777, 1036, 1295, 3990, 4218, 24605, 42180, 98420, 98420, 147630, 147630, 147630],$$

because $p = 37 > 2r + 1 = 27$.

Here is a stronger version in the integral case (shared by Eric Rowell and Andrew Schopieray):

Theorem 8.5. *For an integral modular fusion category, for every prime p dividing the global FPdim, there is a basic FPdim of multiplicity m such that $p \leq 2m + 1$.*

Proof. Consider the orbit $(\sigma^2(t_X))$ with at least $(p-1)/2$ distinct elements from above proof of Theorem 8.4. By (3), the orbit $(\sigma(X))$ has also at least $(p-1)/2$ distinct elements. By applying (1) on the (weakly) integral case, we get that $\sigma(\text{FPdim}(X)) = \text{FPdim}(X)$. Thus all simple objects in the orbit $(\sigma(X))$ has the same FPdim, so the multiplicity m of this basic FPdim satisfies $m \geq (p-1)/2$, i.e., $p \leq 2m + 1$. $\square$

Here is an example of type of rank 13 and $\text{FPdim} = 2^4 3^6 5^2 7^2 17^2$ excluded by Theorem 8.5 (but not Theorem 8.4):

$$[1, 238, 459, 540, 595, 918, 5355, 9180, 21420, 21420, 32130, 32130, 32130],$$

because $p = 17 > 2m + 1 = 7$, where $m = 3$ is the largest multiplicity of a basic FPdim.

Remark 8.6. As mentioned by an anonymous referee, a refined version of Theorem 8.5 follows from [5, Lemma 4.6]: in a modular fusion category, for every odd prime p dividing $\text{ord}(t)$, there exists a basic dim with Galois multiplicity (that is, the number of Galois-conjugate basic dims) $m \geq \varphi(p^a)/2 \geq (p-1)/2$, where $a = v_p(\text{ord}(t))$.

Here is a stronger version of Theorem 8.5:

Theorem 8.7. *For an integral modular fusion category, let S be the set of odd prime factors of the global FPdim. There is a partition (S_i) of S , and multiplicities (m_i) of some distinct basic FPdims such that*

$$m_i \geq \frac{1}{2} \text{lcm}_{p \in S_i}(p-1).$$

Proof. Let S_X be the set of odd prime divisors of $\text{ord}(t_X)$. By Cauchy theorem in [7], the union of the sets S_X over all the simple objects X is exactly S . Let λ be the *Carmichael function*, i.e. the exponent of the multiplicative group of integers modulo n . It is well-known that if $n = \prod_i p_i^{n_i}$ is the prime factorization of n , then $\lambda(n) = \text{lcm}_i \lambda(p_i^{n_i})$, whereas for p_i odd then $\lambda(p_i^{n_i}) = \varphi(p_i^{n_i}) = (p_i - 1)p_i^{n_i-1}$, where φ is the Euler totient function. The orbit $(\sigma(X))$ has at least $\frac{1}{2} \lambda(\prod_{p \in S_X} p)$ distinct elements. Therefore, by above and the proof of Theorem 8.5, the multiplicity of $\text{FPdim}(X)$ is at least $\frac{1}{2} \text{lcm}_{p \in S_X}(p-1)$. The result follows. $\square$

Here is an example of type of rank 25 and $\text{FPdim} = 3^4 5^2 7^2 11^2 13^2$ excluded by Theorem 8.7 (but not Theorem 8.5):

$$[[1, 1], [39, 2], [231, 2], [273, 2], [1001, 2], [1287, 2], [3465, 2], [4095, 2], [9009, 2], [15015, 8]].$$

First, note that it is not excluded by Theorem 8.5, since $m = 8 \geq (p-1)/2$ for all $p \in \{3, 5, 7, 11, 13\}$. Next, let us explain why it is excluded by Theorem 8.7. Let S_i and S_j be the components of the partition containing 11 and 13, respectively. The multiplicity $m = 8$ is the only one larger than $(p-1)/2$ for $p = 11, 13$. Hence $S_i = S_j$, but $\text{lcm}(13-1, 11-1)/2 = 30 > 8$, a contradiction.

The checking of Theorems 8.5 and 8.7 are automated by the function `Theorem8_5Check` and `Theorem8_7Check` in `TypeCriteria.sage` in [49].

9. STRONGER ARITHMETIC CONSTRAINTS

In this section, we will first prove some arithmetic constraints on the rank of the Drinfeld center of $\text{Rep}(G)$, for any finite group G . Then, we will discuss how far they can be generalized to any integral modular fusion categories.

9.1. Rank of $\mathcal{Z}(\text{Rep}(G))$. This subsection is inspired from discussions with Geoff Robinson and Dave Benson in [46, 47]. The goal is to simplify the group theoretic way to express the rank of $\mathcal{Z}(\text{Rep}(G))$, for any finite group G , and to provide some bounds involving the prime divisors of $|G|$. Consequently, this class of integral modular fusion categories satisfies Conjectures 9.8 and 9.11. We will adopt the following notations throughout our discussion:

- $Z(G)$ denotes the center of G ,
- Γ_G represents a complete set of conjugacy class representatives,
- c_G is the rank of $\text{Rep}(G)$, which equals the total number of conjugacy classes in G , denoted by $|\Gamma_G|$,
- r_G refers to the rank of $\mathcal{Z}(\text{Rep}(G))$.

According to [14] or [36, Section 3], the rank r_G is determined by the number of irreducible characters in the centralizers of class representatives of G , as expressed precisely in Equation (9.1) below.

Lemma 9.1. *Let G be a finite group. Then*

$$(9.1) \quad r_G = \sum_{a \in \Gamma_G} c_{C_G(a)}.$$

The following result provides a simpler expression for the rank r_G . An anonymous referee pointed out that it is already stated in the expository paper [34, Corollary 2.3].

Lemma 9.2. *The rank r_G is the number of conjugacy classes of ordered pairs of commuting elements of G .*

Proof. By the equality in Lemma 9.1, it suffices to establish a bijection between the set

$$A_G := \{c(a_1, a_2) \mid a_1, a_2 \in G \text{ with } a_1 a_2 = a_2 a_1\},$$

where

$$c(a_1, a_2) := \{(ga_1 g^{-1}, ga_2 g^{-1}) \mid g \in G\},$$

and the set

$$B_G := \{(a, \beta) \mid a \in \Gamma_G \text{ and } \beta \text{ is a conjugacy class within } C_G(a)\}.$$

Given $c(a_1, a_2) \in A_G$, we associate the element $(a_1, \{ha_2 h^{-1} \mid h \in C_G(a_1)\}) \in B_G$. We merely need to confirm that if $a_1 = ga_1 g^{-1}$, then a_2 and $ga_2 g^{-1}$ are conjugates in $C_G(a_1)$, which is apparent since $a_1 = ga_1 g^{-1}$ means that $g \in C_G(a_1)$. Given $(a, \beta) \in B_G$, we associate the element $c(a, b) \in A_G$, where $b \in \beta$. We only need to verify that if $b' \in \beta$, then $c(a, b') = c(a, b)$. Note that $b' = h b h^{-1}$, where $h \in C_G(a)$. Therefore, $c(a, b) = c(h a h^{-1}, h b h^{-1}) = c(a, b')$ because $h a h^{-1} = a$, given that $h \in C_G(a)$. $\square$

The following Proposition 9.3, Theorem 9.5, and Theorem 9.6 form a sequence of increasingly stronger results, each proven independently. We include all the proofs, rather than only the final one (see §A), due to the significant increase in complexity. Recall that a prime p divides $\text{FPdim}(\mathcal{Z}(\text{Rep}(G))) = |G|^2$ if and only if it divides $|G|$.

Proposition 9.3. *Let G be a finite group. Let p be a prime divisor of $|G|$. Then $r_G \geq p$.*

Proof. This proof is due to Dave Benson. The ordered pair of commuting elements (g, g^i) for $1 \leq i \leq \text{ord}(g)$ are all in distinct conjugacy classes, so by Lemma 9.2, $r_G \geq \text{ord}(g)$ for all g in G . By Cauchy's theorem, there is an element of order p dividing $|G|$. Thus, $r_G \geq p$. $\square$

The number of conjugacy classes of ordered pairs of commuting elements in the alternating group A_n is 1, 1, 9, 14, 22, 44, 74 for $n = 1, \dots, 7$, respectively, see [48].

Lemma 9.4. *Let G be a finite group. Then*

$$(9.2) \quad r_G \geq |Z(G)|c_G + \sum_{g \in \Gamma_G \setminus Z(G)} \text{ord}(g),$$

$$(9.3) \quad r_G \geq \sum_{g \in \Gamma_G} \text{ord}(g).$$

Proof. In general, we have

$$(9.4) \quad c_{C_G(a)} \geq |Z(C_G(a))| \geq \text{ord}(a)$$

but if $a \in Z(G)$ then $C_G(a) = G$ and so $c_{C_G(a)} = c_G$. The inequalities (9.2) and (9.3) follow from (9.1). $\square$

Theorem 9.5. *Let G be a finite group. Let p be a prime divisor of $|G|$. Then $r_G \geq 2p$.*

Proof. By Cauchy's theorem, there is an element g of order p . If $p = 2$ then $r_G \geq 3$ by (9.3). So if $r_G \leq 3$, then $r_G = 3$ and $|\Gamma_G| = 2$ by (9.3), thus $G = C_2$, contradiction. So $r_G \geq 4 = 2p$.

Therefore, we can assume that p is odd. If g^2 is not in the conjugacy class of g then, by (9.1) and (9.4),

$$r_G \geq c_{C_G(g)} + c_{C_G(g^2)} \geq |Z(C_G(g))| + |Z(C_G(g^2))| \geq 2p,$$

because $x \in Z(C_G(x))$ and $\text{ord}(g^2) = p$, as p is odd. Thus, we can assume the existence of h in G such that $hgh^{-1} = g^2$, but then $h^n g h^{-n} = g^{2^n}$. Fermat's little theorem states that $2^{p-1} \equiv 1 \pmod{p}$, and the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$. So $h^n g h^{-n} = g$ for $n = p-1$, and $\{h^n g h^{-n} \mid n = 1, \dots, p-1\} = \langle g \rangle \setminus \{e\}$. Thus $p-1$ divides $\text{ord}(h)$. But g and h have different order, so cannot be in the same conjugacy class, so by (9.3), $r_G \geq \text{ord}(e) + \text{ord}(g) + \text{ord}(h) \geq 1 + p + p-1 = 2p$. $\square$

The equality $r_G = 2p$ is realized by $(G, p) = (C_2, 2)$. Out of this example, we can get even better:

Theorem 9.6. *Let G be a finite group. Let p be a prime divisor of $|G|$. Then $r_G \geq 3p - 1 - \delta_{G, C_2}$.*

Proof. If G is Abelian, then by (9.2), we have $r_G = |G|^2$. This means that $r_G < 3p - 1$ implies $G = C_2$, where $r_G = 4 = 3p - 2$. The non-Abelian case, attributed to Geoff Robinson, is much more complex and too lengthy to include here. It is presented in §A. $\square$

The equality $r_G = 3p - 1$ is realized by $(G, p) = (S_3, 3)$. We can expect even better in the non-solvable case:

Conjecture 9.7. *For every non-solvable finite group G , and for every prime p dividing the order of G , the inequality $r_G \geq 5p - 3$ holds.*

This has been verified for all non-solvable groups of order less than 1920 and for all non-Abelian finite simple groups of order less than 10^8 . The equality is achieved for $G = A_5$ and $\text{PSL}(2, 7)$. Geoff Robinson has shown in [47] that $r \geq 5p - 1$ for any non-solvable groups without a self-centralizing Sylow p -subgroup of order p , where p the largest prime factor of $|G|$.

9.2. Integral Modular Fusion Categories. Referring to the notations used previously, recall that Theorem 8.4 asserts that $p \leq 2r + 1$, which is enhanced for the integral case by Theorem 8.5 to $p \leq 2m + 1$, and also consult Theorem 8.7 for a more robust version. Now, let's explore the extent to which the arithmetic constraints discussed in §9.1 can be applied to integral modular fusion categories. The subsequent conjecture aims to expand upon Proposition 9.3.

Conjecture 9.8. *For any prime number p that divides the global dimension of a integral modular fusion category with rank r , then $p \leq r$.*

Proposition 9.9. *The statement of Conjecture 9.8 is true up to rank 21.*

Proof. The proof largely relies on computer assistance. All computational steps are documented in `InvestUpToRank21par.txt` in [49]. The script `all_rep_above_r`, mentioned in §4, focuses on types that comply with Theorem 8.7 yet contradict Conjecture 9.8. Consequently, we identified exactly 187 potential types up to rank 21, as listed in `UpToRank21par.txt` in [49]. Specifically, we found none up to rank 13, and then 1, 4, 22, 2, 28, 0, 8, 122 types at ranks 14 through 21, respectively. Of these, only 40 types pass `TypeCriteria` from §5. Among these, only 28 types pass `GradingCriteria` from §8.1. They are ultimately excluded by our fusion ring solver's partition version in §6.4, limited to one second per type. $\square$

The extension of Proposition 9.9 is currently underway, as detailed in `InvestAboveRank21par.txt`. At ranks 22, 23, and 24, there are still 22, 1, and 16 types respectively remaining to be considered; all of them being perfect, thus:

Proposition 9.10. *The statement of Conjecture 9.8 is true up to rank 24 in the non-perfect case.*

If Conjecture 9.8 is true then it is optimal as demonstrated by the pointed examples of prime rank. Thus, we could expect better for the non-pointed case. The following conjecture generalizes Theorem 9.5.

Conjecture 9.11. *For any prime number p that divides the global dimension of a non-pointed integral modular fusion category with rank r , then $p \leq r/2$.*

Proposition 9.12. *The statement of Conjecture 9.11 is true up to rank 15.*

Proof. The proof is mainly computer-assisted, but computationally a bit harder than for Proposition 9.9. All computational steps are documented in `InvestUpToRank15pahr.txt` in [49]. The specialized script `all_rep_above_half_r`, mentioned in §4, provides 3094 non-pointed types up to rank 15 satisfying Theorem 8.7 but contradicting Conjecture 9.11, available in `UpToRank15pahr.txt`. We found none up to rank 8, and then 6, 36, 250, 2266, 45, 491 types at ranks 9 through 15, respectively. The irregularity between rank 13 and rank 14 comes from the fact that if $p > 14/2 = 7$, then $p \geq 11$ because 9 is not prime. Of these, only 1256 types pass `TypeCriteria` from §5. Among these, only 900 types pass `GradingCriteria` from §8.1. The use of our fusion ring solver's partition version from §6.4, limited to one second per type, reduced the rest to just 15 types, then 12, 5, 4 ones with 10, 100, 1000 seconds per type respectively. Next, we applied our full fusion ring solver's partition version from §6.3, requiring to specify the duality, so our 4 types became 40 cases, we simplify with the commutativity option. After the first round, limited to 10 seconds per case, there remain 8 cases from 2 types, ultimately all excluded in about two hours. $\square$

The extension of Proposition 9.12 is currently in progress, as detailed in `InvestAboveRank15pahr.txt`. At ranks 16 and 17, there remain 132 types (all perfect) and 7737 types (of which only 19 are not perfect), respectively, still to be considered. Thus:

Proposition 9.13. *The statement of Conjecture 9.8 is true up to rank 16 in the non-perfect case.*

With our current knowledge, it would not be reasonable to conjecture a generalization of Theorem 9.6 suggesting that $p \leq (r + 1)/3$ in the non-pointed case, or of Conjecture 9.7 suggesting that $p \leq (r + 3)/5$ in the non-solvable case. But it makes sense to explore in this direction.

Question 9.14. *Is it true that for any prime number p dividing the global dimension of a non-solvable integral modular fusion category with rank r , the inequality $p \leq (r + 3)/5$ holds?*

The following proposition is related to Question 7.1.

Proposition 9.15. *An affirmative response to Question 9.14 would indicate that the minimum rank required for a non-trivial perfect integral modular fusion category is 22.*

Proof. By [24, Proposition 4.5 (iv)], a non-trivial perfect integral fusion category $\mathcal{C}$ is non-solvable. Thus by [24, Theorem 1.6], $\text{FPdim}(\mathcal{C})$ must have at least three distinct prime factors, so there must be a prime factor $p \geq 5$. Assume that $\mathcal{C}$ is modular of rank r , then an affirmative response to Question 9.14 implies that $r \geq 5p - 3 \geq 22$. Finally, the bound is realized by $\mathcal{Z}(\text{Rep}(A_5))$ having rank 22. $\square$

Here is what we can deduce in the perfect integral case:

Corollary 9.16. *For any prime number p that divides the global dimension of a non-trivial perfect integral modular fusion category with rank r , then $p \leq 2r - 5$.*

Proof. By Theorem 5.1, the number of distinct basic FPdims is at least 4. Let p be the biggest prime divisor of the global dimension, then by Theorem 8.5, there is a basic FPdim of multiplicity $m \geq (p-1)/2$. Thus $r \geq m+3 = (p+5)/2$. The result follows. $\square$

10. PROOF UP TO RANK 13

This section proves Theorem 1.3 by leveraging the advanced results from Sections 8 and 9, complemented by computational assistance. All computational steps are documented in `InvestUpToRank13.txt`, as referenced in [49]. The initial step involves classifying all the possible types (4308 in total) constrained by Theorem 8.7, using `all_rep_th87` from §4. Applying Proposition 9.12 reduces this number to just 1722 types. From these, merely 794 types met the `TypeCriteria` outlined in §5. Among these, only 439 types fulfilled the `GradingCriteria` presented in §8.1. Utilizing the partition version of our fusion ring solver, as discussed in §6.4 and limiting the processing time to one second per type, further reduced the count to just 38 types. Subsequently, excluding the type $[1, 1, 5, 5, 6, 6, 20, 24, 30, 40, 60, 60, 60]$ in under 1000 seconds brought the total down to 37 types.

The distribution of these types across ranks 7 to 13 is as follows: 1, 1, 2, 2, 4, 7, 20, respectively. Out of these, 27 types are non-perfect, and 10 are perfect, with each perfect type containing at least one prime-power entry, thereby providing direct proof for the simple case within Theorem 1.5. Following this, we employed the full version of our fusion ring solver, as outlined in §6.3, which necessitated specifying duality. This adjustment resulted in the expansion of our 37 types into 572 cases. We streamlined these cases using the commutativity and `FusionData` options, as elucidated in §6.1. After an initial round of processing, limited to 10 seconds per case, 56 cases remained unresolved, plus 81 cases with solutions. Further rounds of processing, with limits set to 100 seconds and then 1000 seconds, resulted in $29 + 101$ and $23 + 106$ cases, respectively. Ultimately, leveraging HPC capabilities allowed for the resolution of all cases, concluding with solutions for 120 cases across 26 distinct types, as listed below:

- Rank 7: $[1, 1, 1, 1, 2, 2, 2]$,
- Rank 8: $[1, 1, 2, 2, 2, 2, 3, 3]$,
- Rank 9: $[1, 1, 1, 1, 4, 4, 6, 6, 6]$, $[1, 1, 2, 2, 2, 2, 3, 3, 6]$,
- Rank 10: $[1, 1, 1, 2, 2, 2, 2, 2, 3]$, $[1, 1, 2, 3, 3, 4, 4, 4, 6, 6]$,
- Rank 11: $[1, 1, 1, 1, 2, 2, 2, 2, 2, 2]$, $[1, 1, 1, 1, 2, 4, 4, 4, 4, 6, 6]$, $[1, 1, 1, 1, 4, 4, 12, 12, 18, 18, 18]$, $[1, 1, 1, 3, 4, 4, 4, 4, 4, 6]$,
- Rank 12: $[1, 1, 1, 1, 3, 3, 3, 3, 4, 4, 6, 6]$, $[1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3]$, $[1, 1, 2, 2, 2, 2, 3, 3, 6, 6, 6, 6]$, $[1, 1, 2, 2, 2, 2, 6, 6, 6, 6, 9, 9]$, $[1, 1, 2, 3, 3, 6, 6, 8, 8, 8, 12, 12]$, $[1, 1, 2, 6, 6, 6, 6, 10, 10, 10, 15, 15]$,
- Rank 13: $[1, 1, 1, 1, 2, 2, 4, 4, 4, 4, 4, 4, 6]$, $[1, 1, 1, 1, 4, 4, 4, 4, 4, 4, 10, 10, 10]$, $[1, 1, 1, 1, 4, 4, 6, 6, 6, 6, 6, 6, 6]$, $[1, 1, 1, 1, 4, 4, 6, 12, 12, 12, 12, 18, 18]$, $[1, 1, 1, 1, 4, 4, 12, 12, 36, 36, 54, 54, 54]$, $[1, 1, 1, 3, 12, 12, 20, 20, 20, 20, 20, 20, 30]$, $[1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 6]$, $[1, 1, 2, 2, 2, 2, 3, 3, 6, 6, 6, 6, 6]$, $[1, 1, 2, 2, 2, 2, 6, 6, 6, 6, 9, 9, 18]$, $[1, 1, 2, 3, 3, 24, 30, 30, 40, 40, 40, 60, 60]$.

Note that none are perfect, which concludes the proof of Theorem 1.5 at this stage. The total number of commutative fusion rings obtained is 30084. Subsequently, the function `ModularGrading` in `ModularData.sage` narrows this down to 8720 fusion rings that have a modular grading (refer to Definition 8.1), across 14 types (a new feature in `Normaliz` allows for the direct classification of fusion rings with modular grading; see [12, §H.5.2]). However, only 42 fusion rings (spanning 10 types) are both cyclotomic and self-transposable (refer to §3.1), distributed as 4, 0, 6, 6, 16, 10 across ranks 8 to 13. These are detailed in `FusionRingsCCSTMG.sage`. The subsequent use of `MagicCriterion` from §3.2 further refines the list to 13 fusion rings (across 5 types), distributed as 4, 0, 2, 4, 6, 0. Notably, this stage confirms the absence of any non-pointed integral modular fusion categories of rank 13. Lastly, applying `STmatrix` narrows it down to 3 (non-pointed) types encompassing 5 fusion rings and 19 modular data (MD):

- $[1, 1, 2, 2, 2, 2, 3, 3]$ with 8 MD,
- $[1, 1, 1, 2, 2, 2, 2, 2, 2, 3]$ with 3 MD,
- $[1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2]$ with 8 MD,

The pointed cases were addressed by using `STmatrixCo` on a list of fusion rings for the abelian groups of order up to 13. We concluded with $19 + 64$ sets of modular data, originating from $5 + 18$ fusion rings representing $3 + 13$

types (non-pointed + pointed), up to rank 13. This proves Theorem 1.3. The full details are available in [3, §I] or `MDuptoRank13.txt` at [49].

Remark 10.1. When considering isomorphism classes, it is appropriate to adopt a *normal form* by sorting the basic elements according to their `FPdim` and topological spin, and limit basis permutations to those preserving both.

The `STmatrix` function ran quickly on all remaining fusion rings, with the exception of two of the following type which took several hours to be excluded: $[1, 1, 2, 2, 2, 2, 6, 6, 6, 9, 9, 18]$.

11. ABOUT RANKS 14 AND 15

The current state of exploration at ranks 14 and 15 is documented in `InvestRank14.txt` and `InvestRank15.txt`, in [49]. Regarding rank 14, the initial count was 29113 (non-pointed) types, constrained by Theorem 8.7. At the time of writing, only 35 types remain to be considered, 8 of which are non-perfect. This leads to the following propositions:

Proposition 11.1. *The type of a non-perfect non-pointed integral modular fusion category of rank 14 belongs to the following list:*

- $[1, 1, 2, 3, 3, 24, 24, 42, 42, 56, 56, 56, 84, 84]$,
- $[1, 1, 2, 3, 3, 24, 120, 150, 150, 200, 200, 200, 300, 300]$,
- $[1, 1, 24, 24, 36, 40, 45, 45, 90, 90, 90, 180, 180, 180]$,
- $[1, 1, 40, 84, 90, 126, 315, 315, 504, 630, 840, 1260, 1260, 1260]$,
- $[1, 1, 45, 45, 90, 140, 168, 168, 630, 630, 840, 1260, 1260, 1260]$,
- $[1, 1, 60, 60, 84, 140, 189, 189, 540, 1260, 1260, 1890, 1890, 1890]$,
- $[1, 1, 90, 90, 90, 108, 140, 378, 945, 945, 1260, 1890, 1890, 1890]$,

together with

- $[1, 1, 1, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4]$

which admits four commutative cyclotomic self-transposable fusion rings, leading to 6 MD, all from $\mathcal{Z}(\text{Vec}(A_4, \omega))$, already listed in the database of [29] at directory path `/Modular_Data/12/3/`.

For the perfect case, you can consult the list of 27 possible types in `InvestRank14.txt`.

Corollary 11.2. *The set of prime divisors of the global `FPdim` of a non-pointed integral modular fusion category of rank 14 must be:*

- $\{2, 3, 5, 7\}$ in the perfect case,
- $\{2, 3, 5\}$, $\{2, 3, 5, 7\}$, or $\{2, 3, 7\}$ in the non-perfect case, except $\mathcal{Z}(\text{Vec}(A_4, \omega))$ for which it is $\{2, 3\}$.

Regarding rank 15, the initial count was 333423 (non-pointed) types, also constrained by Theorem 8.7. At the time of writing, only 9027 types remain to be considered, of which 399 are non-perfect. The list of possible types are in `InvestRank15.txt`.

Corollary 11.3. *The set of prime divisors of the global `FPdim` of a non-pointed integral modular fusion category of rank 15 must be:*

- $\{2, 3, 5\}$, $\{2, 3, 5, 7\}$, or $\{2, 3, 7\}$ in the perfect case,
- $\{2, 3\}$, $\{2, 3, 5\}$, $\{2, 3, 5, 7\}$, or $\{2, 3, 7\}$ in the non-perfect case.

Remark 11.4. In the subsequent paper [20], a generalization of an argument of Etingof–Nikshych–Ostrik yields a powerful necessary criterion for integral modular categorification. Using this criterion, we complete the classification of categorifiable integral modular data up to rank 14, and up to rank 25 in the odd-dimensional case. We also demonstrate the ongoing effectiveness of this approach by reducing the number of unresolved types at rank 15 from 9027 to just 2481.

12. THE ODD-DIMENSIONAL CASE

For an overview of the current state of knowledge on odd-dimensional modular fusion categories, we refer the reader to [15, 16]. A foundational result in this area establishes that an odd-dimensional modular fusion category $\mathcal{C}$ is equivalent to being maximally non self-dual (MNSD), meaning that its only self-dual simple object is the unit object. Let $(d_i)_{i \in I}$ represent the basic `FPdims` of $\mathcal{C}$. Since d_i^2 is a divisor of the odd `FPdim`($\mathcal{C}$), each d_i must be odd. Furthermore, the equation $\sum_{i \in I} d_i^2 = \text{FPdim}(\mathcal{C})$ implies that the rank $r = |I|$ must also be odd. This reduces our investigation to Egyptian fractions of the form $q = \sum_{i=1}^r \frac{1}{s_i^2}$, where $q, r, s_i \in \mathbb{Z}_{\geq 1}$, $s_1 \geq \dots \geq s_r \geq 1$, and both r and s_i are odd. Additionally, s_i divides s_1 for all i , and $s_{2k} = s_{2k+1}$. This yields the expression

$$q = \frac{1}{s_1^2} + \sum_{k=1}^{(r-1)/2} \frac{2}{s_{2k}^2}.$$

Since each s_i is odd, we have $s_i^2 \equiv 1 \pmod{8}$, which implies $q \equiv r \pmod{8}$ and that q is odd as well. Utilizing a similar technique as in §4, we can assume $s_i > 1$ (hence $s_i \geq 3$), by completing the classification with additional 1s if necessary. Consequently, we can assume $q \leq r/9$. For $r < 27$, this allows us to deduce that $q = 1$, and therefore $r \equiv 1 \pmod{8}$, which narrows the possibilities for r to 1, 9, 17, 25 (up to completing by 1s).

Remark 12.1. This strategy can be extended. For instance, by adding eighteen 3s to complete the classification, we may assume that $s_i \geq 5$ for $i \leq r - 16$, which leads to $q \leq 16/9 + (r - 16)/25$. If $r < 47$ (which becomes 51 because $q \equiv r \pmod{8}$), we can assume that $q = 1$. However, this extended strategy will not be applied in this paper.

For all $r < 25$, we compiled the following list of all possible non-pointed types (using `all_rep_MNSD` from §4):

- $[[1, 9], [3, 8], [9, 2a]]$,
- $[[1, 7], [3, 2], [5, 8], [15, 2a]]$,
- $[[1, 3], [3, 8], [5, 6], [15, 2a]]$,
- $[[1, 1], [3, 2], [7, 2], [9, 4], [21, 8], [63, 2a]]$,
- $[[1, 1], [9, 4], [25, 2], [45, 2], [75, 8], [225, 2a]]$,

where $a \geq 0$ represents the number of 1s added for completion. It is noteworthy that these ranks are $17 + 2a$, which corroborates a result from [15] stating that any odd-dimensional modular fusion category with rank less than 17 is pointed. Further, [15, Remark 4.3] states that any perfect odd-dimensional modular fusion category is a Deligne product of simple ones. From the preceding analysis, a non-pointed one must have a rank of at least 17, meaning a perfect non-simple one must have a rank of at least $289 (= 17^2)$. Therefore, a perfect one with a rank less than 289 must be simple, and cannot have a prime-power basic FPdim, as shown in [43, Corollary 6.16]. Consequently, the previously mentioned perfect types are excluded. It follows that:

Theorem 12.2. *Every perfect odd-dimensional modular fusion category of rank less than 25 is trivial, and so everyone of rank less than 625 ($= 25^2$) is simple.*

Proof of Theorem 1.9

Proof. By Theorem 12.2, there remain to address the non-perfect types above. Their rank is always $17 + 2a < 25$, which implies $a < 4$. As outlined in §8.1, the modular grading results in a partition indexed by the pointed part, with each component having the same FPdim, in particular the FPdim of the pointed part divides the global FPdim.

- First, let's examine the type $[[1, 9], [3, 8], [9, 2a]]$. The FPdim for this type is $81(1 + 2a)$. Consequently, each partition component must have $\text{FPdim} = 9(1 + 2a)$. If $a > 0$, a component with 9 must have $\text{FPdim} \geq 81$. This leads to $81 \leq 9(1 + 2a)$, resulting in $a \geq 4$, contradiction. Thus, $a = 0$. The modular partition must be

$$[[1, 1, 1, 1, 1, 1, 1, 1], [3], [3], [3], [3], [3], [3], [3], [3]],$$

which contradicts Theorem 8.2 (1).

- Regarding the second type $[[1, 7], [3, 2], [5, 8], [15, 2a]]$, the FPdim of the pointed part equaling 7 is not a divisor of the global $\text{FPdim} = 225(1 + 2a)$, for $0 \leq a < 4$, except $a = 3$. Therefore, each partition component must have $\text{FPdim} = 225(1 + 2 \times 3)/7 = 225 = 15^2$, so the modular partition must be

$$[[1, 1, 1, 1, 1, 1, 1, 3, 3, 5, 5, 5, 5, 5, 5, 5], [15], [15], [15], [15], [15], [15]],$$

which contradicts Theorem 8.2 (1).

- Lastly, consider the third type $[[1, 3], [3, 8], [5, 6], [15, 2a]]$. If $a = 1$ then the modular partition must be

$$[[1, 1, 1, 3, 3, 3, 3, 3, 3, 3, 5, 5, 5, 5, 5, 5], [15], [15]],$$

which contradicts Theorem 8.2 (1). If $a = 2$, then there is no modular partition. If $a = 3$, then the global $\text{FPdim} = 3^2 5^2 7$, but its powerless prime factor $p = 7$ does not divide $\text{FPdim}(\mathcal{C}_{pt}) = 3$, contradicting [13, Theorem 1.17]. Alternatively, see Remark 12.3. Hence, $a = 0$. Following this, applying the fusion ring solver outlined in §6.3 to the type $[[1, 3], [3, 8], [5, 6]]$ yields two fusion rings. Applying `STmatrix2` to these fusion rings provides 3 modular data, detailed in [3, §II]. $\square$

Remark 12.3. Here is an alternative proof for the case $a = 3$ in the last item above. The neutral component $\mathcal{C}_e$ must be of type $[[1, 3], [3, 8], [15, 2]]$ and $\text{FPdim} = 3^1 5^2 7$, so its modularization $\mathcal{M}$ would have $\text{FPdim} = 5^2 7$, hence cannot have basic $\text{FPdim} = 3$, thus must be of type $[[1, 25], [5, 6]]$, but each component of the modular partition (for $\mathcal{M}$) would have $\text{FPdim} = 7$, and so $5^2 \leq 7$, contradiction.

Definition 12.4. A modular data is called *anomaly-free* if its Gauss sums are equal ($p_+ = p_-$), see Definition 2.11.

Lemma 12.5. *A modular data is anomaly-free if and only if $p_+ = \pm \sqrt{\dim}$, if and only if the central charge $c \in \{0, 4\}$.*

Proof. Recall from Definition 2.11 that $p_{\pm} := \sum_{i=1}^r d_i^2(\theta_i)^{\pm 1}$. Thus p_+ and p_- are complex-conjugate. So anomaly-free is equivalent to p_+ real. Now, $p_+ = \sqrt{\dim} \zeta_g^c$, thus it is real if and only if $\zeta_g^c = \pm 1$, if and only if $c = 0$ or 4. $\square$

Remark 12.6. As highlighted in Remark 1.10, gaps have been identified in the literature:

- (1) In [6, Theorem 4.2, proof of Case (viii) $\text{FPdim}(\mathcal{C}_{pt}) = p$], on page 727, the assertion that the anomaly-freeness (as defined in their reference [21], see Definition 12.4) necessarily leads to $p_+ = pq$ is incorrect, see Lemma 12.5, it may also be $-pq$, as for the MD described in [3, §II], thus also allowing $p|(q+1)$.
- (2) In [15, Theorem 6.3 (b), proof of Case $|\mathcal{G}(\mathcal{C})| = 3$], on page 1936, the deduction “Hence $l \leq 24$ ” in the seventh last line is accurate, except when $c_{X_1} = 1$, which permits $l = 5$, thereby accommodating the type $[[1, 3], [3, 8], [5, 6]]$.

Following our paper, [15] was corrected on arXiv, and [26] introduces modular categorifications for these new MD.

Remark 12.7. Regarding the modular categorifications $\mathcal{C}$ of these new MD in [3, §II], a discussion with Sebastian Burciu revealed that the Grothendieck ring of the braided adjoint fusion subcategory $\mathcal{D} = \mathcal{C}_{ad}$, which is of rank 11, FPdim 75, type $[[1, 3], [3, 8]]$, and basis $\{b_g\}_{g \in C_3} \cup \{x_i, x_i^*\}_{i \in \{1, 2, 3, 4\}}$, is equal to $\text{ch}(G)$, where $G = C_5^2 \rtimes C_3$ is the unique non-Abelian finite group of order 75. In fact, an application of §6.3 shows a unique MNSD cyclotomic fusion ring of this type. Furthermore, $\mathcal{D}$ is not symmetric, as indicated by the S -matrices mentioned in [3, §II]. In fact, the Müger center $\mathcal{Z}_2(\mathcal{D})$ is $\mathcal{C}_{pt}$, pointed of rank 3. Therefore, $\mathcal{D}$ can be $\text{Rep}(G)$, albeit with an unusual braiding (see [18]), or, more broadly, a Jordan-Larson category [31] with an FPdim of 3×5^2 . Finally, according to the S -matrices again, $\mathcal{C}$ is a minimal modular extension of $\mathcal{D}$, see [35] and [30, §1.1].

There are non-pointed and non-perfect odd-dimensional modular fusion categories of rank 25, exemplified by $\mathcal{Z}(\text{Vec}_{C_7 \rtimes C_3}^\omega)$. Furthermore, [16] demonstrates that, up to equivalence, no additional examples exist. Consequently, our attention must now turn to the examination of the perfect case.

Proposition 12.8. *A perfect odd-dimensional modular fusion category of rank 25, if any, must have one of the following 3 types:*

1. $[[1, 1], [75, 2], [91, 4], [175, 2], [585, 2], [975, 2], [2275, 2], [4095, 2], [6825, 8]]$,
2. $[[1, 1], [75, 2], [91, 4], [175, 2], [975, 2], [2275, 2], [2925, 4], [6825, 8]]$,
3. $[[1, 1], [135, 4], [165, 2], [189, 2], [315, 2], [385, 2], [1155, 2], [2079, 2], [3465, 8]]$.

Proof. We are reduced to the simple case by Theorem 12.2. Moreover, $q = 1$; otherwise, the type would be a completion (via the process of adding 1s to the Egyptian fraction with squared denominators) of a perfect type with a rank less than 25. However, as previously mentioned, all such types possess some entries that are prime-power, so they do not conform to the simple case as outlined in [43, Corollary 6.16]. The investigation is documented in `InvestMNSDPerfectR25.txt`. We started with 91 possible types using the aforementioned method combined with [43, Corollary 6.16]. Among them, only 15 types pass `Theorem8_7Check` (i.e., Theorem 8.7), and then 7 pass `TypesCriteria`. Using our fusion ring solver in §6.4 and HPC, we reduced the list to the 3 mentioned types. $\square$

These three types are resolved in a subsequent paper; see Remark 11.4.

A. PROOF BY GEOFF ROBINSON

This proof, originally due to Geoff Robinson [47] and slightly simplified here, completes the argument for Theorem 9.6 by establishing that $r_G \geq 3p - 1$ when G is non-Abelian. Most of the steps are straightforward, though the final part relies on a deeper result from modular representation theory, by R. Brauer [4].

- (1) We may assume that p is the largest prime dividing $|G|$; if not, we replace p with the largest one.
- (2) The number of conjugacy classes of the centralizer $C_G(a)$ is at least the order of its center $Z(C_G(a))$. Furthermore, since $Z(C_G(a))$ contains both $Z(G)$ and $\langle a \rangle$, we have:

$$|Z(C_G(a))| \geq \text{lcm}(\text{ord}(a), |Z(G)|).$$

From this point onward, assume that $r_G \leq 3p - 2$. Under this condition, there can be at most two non-conjugate elements of order p , and no element g of order kp for any integer $k > 1$. The existence of such an element would imply

$$r_G \geq \text{ord}(g) + \text{ord}(g^k) = kp + p = (k+1)p \geq 3p > 3p - 2,$$

since two elements with different order, like g and g^k , cannot be conjugate. This contradicts the assumption.

- (3) Let $x \in G$ be an element of order p . Suppose $C_G(x)$ is not a p -group. Then it contains an element y of prime order $q \neq p$. Since x and y commute and their orders are coprime, the element xy has order pq , contradicting the bound in Step 2 since $q > 1$. Therefore, $C_G(x)$ must be a p -group.
- (4) **Lemma:** Let H be a non-cyclic p -group. Then H contains at least $3p - 2$ conjugacy classes.

Proof. If H is abelian, then it must have order p^n with $n \geq 2$ (since it is non-cyclic), and the number of conjugacy classes equals $|H| \geq p^2 \geq 3p - 2$.

Recall that a group of order p^2 must be Abelian. Suppose H is non-abelian. Then $|H| = p^n$ with $n \geq 3$. Let $g_1, \dots, g_m$ be representatives of the distinct conjugacy classes in H . By the class equation,

$$|H| = \sum_{i=1}^m |H : C_H(g_i)|.$$

One term in the sum is 1, corresponding to the neutral element. Since all the indices $|H : C_H(g_i)|$ are powers of p , and the sum is congruent to 1 mod p , there are at least p terms equal to 1.

The remaining $m - p$ terms sum to $|H| - p$. Modulo p^2 , we conclude that at least $p - 1$ of these terms must be at most p . Therefore, we have at least $2p - 1$ terms each at most p , and their total contribution is at most:

$$p + (p - 1)p = p^2.$$

Since $|H| = p^n$ with $n \geq 3$, modulo p^3 , we get that at least $p - 1$ additional terms must be at most p^2 . Hence, altogether, there are at least $3p - 2$ conjugacy classes. $\square$

- (5) If $C_G(x)$ is not cyclic, then by Step 4, it has at least $3p - 2$ conjugacy classes. Since $C_G(e) = G$ has at least 2 conjugacy classes, we obtain $r_G \geq 3p$, which contradicts the assumption. Hence, $C_G(x) = \langle x \rangle$ for every $x \in G$ of order p .
- (6) Let H be a Sylow p -subgroup of G . If $|H| = p^n$ with $n > 1$, by the class equation modulo p , $|Z(H)| = p^m$ with $m \geq 1$, so H has a central element x of order p , and $C_G(x)$ contains H , contradicting Step 5.
- (7) By Step 6, G has a cyclic Sylow p -subgroup of order p . If $p = 2$, then by Step 1, $G \cong C_2$, but G is non-Abelian. Therefore, p is odd, and the Sylow p -subgroup of G is $H = \langle x \rangle = C_G(x)$, of order p .
- (8) Let $G_1 = \{g \in G : gH = Hg\}$ be the normalizer of H in G . Now, H is a normal (and hence unique) Sylow p -subgroup of G_1 , as all Sylow subgroups are conjugate. Let $|G_1| = ep$, with e and p coprime. Brauer's result [4] establishes that the entire group G — not merely G_1 — has at least $e + \frac{p-1}{e}$ conjugacy classes. Moreover, two distinct elements $x_1, x_2 \in H$ are conjugate in G if and only if they are conjugate in G_1 . This is because any $g \in G$ satisfying $gx_1g^{-1} = x_2$ must, by definition, lie in G_1 .

There is an action of G_1 on H by conjugation, yielding a homomorphism

$$\Phi : G_1 \rightarrow \text{Aut}(H) \cong C_{p-1}.$$

The kernel of this homomorphism is H , since $H = C_G(x)$, making H self-centralizing.

Each automorphism $\theta \in \text{Aut}(H)$ is determined uniquely by the image of x , so the number of conjugates of x in H is exactly $|\Phi(G_1)|$, and the same holds for any generator of H . Thus, the index of $\Phi(G_1)$ in $\text{Aut}(H)$ is 1 if all order p elements in G are conjugate; otherwise, it is 2. This follows from Step 2, which shows that there can be at most two conjugacy classes of such elements—and these classes must be of equal size, since the centralizer of any generator of H is always H itself. Since

$$|G_1| = ep = |\Phi(G_1)| \cdot |\ker \Phi| = |\Phi(G_1)| \cdot p,$$

it follows that $|\Phi(G_1)| = e$. Therefore, $\frac{p-1}{e} \in \{1, 2\}$ is the number of conjugacy classes of elements of order p in G . As a subgroup of a cyclic group, $\Phi(G_1)$ is also cyclic, so generated by an element of order e , and G contains an element of order divisible by e .

If $e > 1$, then any element whose order is divisible by e must be nontrivial. Moreover, such an element cannot have order p , since e and p are coprime. Putting everything together, the number of conjugacy classes satisfies:

$$r_G \geq \left(e + \frac{p-1}{e}\right) + e + \frac{p-1}{e}p \geq 3p - 1,$$

- the first term bounds the number of conjugacy classes in $C_G(1) = G$ by Brauer's result,
- the second corresponds to the non-trivial element of order divisible by e ,
- the third accounts for $\frac{p-1}{e}$ conjugacy classes of elements of order p .

Observe that the second inequality follows immediately upon setting $\frac{p-1}{e} = 1$ or 2.

Finally, if $e = 1$, then $p = 3$. Given that p is the largest prime divisor of $|G|$, that 9 does not divide $|G|$, and that G is non-Abelian, it follows that G contains an element of order 2. This element can thus be used in our bound in place of one whose order is divisible by e .

ABOUT THE SUPPLEMENTARY MATERIAL

The complete fusion and modular data are provided separately in the Supplementary Material accompanying this paper, which is cited as [3].

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Availability of data and materials. Data for the computations in this paper are available on reasonable request from the authors. The softwares used for the computations can be downloaded from the URLs listed in the references.

Conflict of interest statement. On behalf of all authors, the corresponding author declares that there are no conflicts of interest.

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